

Thank you for purchasing this Esco Biological Safety Cabinet. Read this manual thoroughly to familiarize yourself with the many unique features and exciting innovations we have built into your new equipment. Esco provides many other resources at our website, www.escoglobal.com, to complement this manual and help you enjoy many years of productive and safe use of your Esco products.



Service Manual

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Airstream[®]

Class II

Biological Safety Cabinet

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“Material in this manual is provided for informational purposes only. The contents and the product described in this manual (including any appendix, addendum, attachment or inclusion), are subject to change without notice. Esco makes no representations or warranties as to the accuracy of the information contained in this manual. In no event shall Esco be held liable for any damages, direct or consequential, arising out of or related to the use of this manual.”

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Warranty Terms and Conditions

Esco products come with a limited warranty. The warranty period will vary depending on the product purchased, beginning on the date of shipment from any Esco international warehouse location. To determine which warranty applies to your product, refer to the appendix below.

Esco's limited warranty covers defects in materials and workmanship. Esco's liability under this limited warranty shall be, at our option, to repair or replace any defective parts of the equipment, provided that these parts, if proven to the satisfaction of Esco, were defective at the time of being sold and that all defective parts shall be returned, properly identified with a Return Authorization.

This limited warranty covers parts only, and not transportation/insurance charges.

This limited warranty does not cover:

- Freight or installation (inside delivery handling) damage. If your product was damaged in transit, you must file a claim directly with the freight carrier.
- Products with missing or defaced serial numbers.
- Products for which Esco has not received payment.
- Problems that result from:
 - External causes such as accident, abuse, misuse, problems with electrical power, improper operating environmental conditions.
 - Servicing not authorized by Esco.
 - Usage that is not in accordance with product instructions.
 - Failure to follow the product instructions.
 - Failure to perform preventive maintenance.
 - Using accessories, parts, or components not supplied by Esco.
 - Damage by fire, floods, or acts of God.
 - Customer modifications to the product.
- Consumables such as filters (HEPA, ULPA, carbon, pre-filters) and fluorescent / UV bulbs.
- Esco is not liable for any damage incurred on the objects used on or stored in Esco equipment. Users are advised to conduct risk assessment and add safety protocols based on their application and sample.

Factory-installed, customer specified equipment or accessories are warranted only to the extent guaranteed by the original manufacturer. The customer agrees that concerning these products purchased through Esco, our limited warranty shall not apply, and the original manufacturer's warranty shall be the sole warranty in respect of these products. The customer shall utilize that warranty for the support of such products and in any event not look to Esco for such warranty support.

Esco encourages all users to register their equipment online at www.escoglobal.com/warranty_registrations.php or complete the warranty registration form included with each product.

ALL EXPRESS AND IMPLIED WARRANTIES FOR THE PRODUCT, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES AND CONDITIONS OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE LIMITED IN TIME TO THE TERM OF THIS LIMITED WARRANTY. NO WARRANTIES, WHETHER EXPRESS OR IMPLIED, WILL APPLY AFTER THE LIMITED WARRANTY PERIOD HAS EXPIRED. ESCO DOES NOT ACCEPT LIABILITY BEYOND THE REMEDIES PROVIDED FOR IN THIS LIMITED WARRANTY OR FOR SPECIAL, INDIRECT, CONSEQUENTIAL OR INCIDENTAL DAMAGES, INCLUDING, WITHOUT LIMITATION, ANY LIABILITY FOR THIRD-PARTY CLAIMS AGAINST YOU FOR DAMAGES, FOR

PRODUCTS NOT BEING AVAILABLE FOR USE, OR FOR LOST WORK. ESCO'S LIABILITY WILL BE NO MORE THAN THE AMOUNT YOU PAID FOR THE PRODUCT THAT IS THE SUBJECT OF A CLAIM. THIS IS THE MAXIMUM AMOUNT FOR WHICH ESCO IS RESPONSIBLE.

These Terms and Conditions shall be governed by and construed in accordance with the laws of Singapore and shall be subject to the exclusive jurisdiction of the courts of Singapore.

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Global Email Helpdesk: support@escoglobal.com

Visit <http://escoglobal.com> to talk to a Live Support Representative

Distributors are encouraged to visit the Distributor Intranet for self-help materials.

Product Appendix, Warranty Listings

Biological Safety Cabinets, Laminar Flow Cabinets, Laboratory Animal Research Workstations, HEPA-Filtered Cabinets (except Streamline brand)	3 years limited
Laboratory Fume Hoods	2 years limited
Ductless Fume Hoods	2 years limited
Cleanroom Equipment	1 year limited
Laboratory Ovens and Incubators	1 year limited
Laboratory Shaker	1 year limited
CO ₂ Incubators	2 years limited
Containment/Pharma Products	2 years limited
Ultra-low Temperature Freezer	5 years limited 5 years on compressor
Laboratory Centrifuge	2 years limited
Freeze Dryer	2 years limited
Laboratory Refrigerator	2 years limited
Thermal Cyclers	3 years limited for MiniPro, Aeris, Provocell 2 years limited for Spectrum 2 years on blocks and Peltier units 2 years on all electronic components 3 years on the housing and fans

Note: The warranty periods may vary by country. Contact your local distributor for specific warranty details.

For international distributors, the warranty period starts two months from the date the equipment is shipped from Esco facility. This allows shipping time so the warranty will go into effect at approximately the same time the equipment is delivered to the user. The warranty protection extends to any subsequent owner during the warranty period. Distributors who stock Esco equipment are allowed an additional four months for delivery and installation, providing the product is registered with Esco. User can register their products online at www.escoglobal.com/warranty or complete the warranty registration form included with each product.

Policy updated on 1st January 2015 (This limited warranty policy applies to products purchased on and after 1st January 2015)

Introduction

1. Products Covered

Esco Class II Biological Safety Cabinet						
Model	Electrical Rating	0.6 meters 2 feet	0.9 meters 3 feet	1.2 meters 4 feet	1.5 meters 5 feet	1.8 meters 6 feet
Airstream	220-240 VAC, 50/60Hz, 1Φ	AC2-2E8	AC2-3E8	AC2-4E8	AC2-5E8	AC2-6E8
		AC2-2S8	AC2-3E8-M	AC2-4E8-M	AC2-5E8-M	AC2-6E8-M
			AC2-3S8	AC2-4S8	AC2-5S8	AC2-6S8
			AC2-3S8-M	AC2-4S8-M	AC2-5S8-M	AC2-6S8-M
Airstream	110-130 VAC, 50/60Hz, 1Φ	AC2-2E9	AC2-3E9	AC2-4E9	AC2-5E9	AC2-6E9
		AC2-2S9	AC2-3E9-M	AC2-4E9-M	AC2-5E9-M	AC2-6E9-M
			AC2-3S9	AC2-4S9	AC2-5S9	AC2-6S9
			AC2-3S9-M	AC2-4S9-M	AC2-5S9-M	AC2-6S9-M
Airstream Duo	220-240 VAC, 50/60Hz, 1Φ			AC2-4D8 AC2-4G8		AC2-6D8 AC2-6G8

2. Safety Warning

- Anyone working with, on or around this equipment should read this manual. Failure to read, understand and comply with the instructions given in this manual may result in damage to the unit, injury to operating personnel, and / or poor equipment performance.
- Any internal adjustment, modification or maintenance to this equipment must be undertaken by qualified service personnel.
- The use of any hazardous materials in this equipment must be monitored by an industrial hygienist, safety officer or some other suitably qualified individuals.
- Before you process, you should thoroughly understand the installation procedures and take note of the environmental / electrical requirements.
- In this manual, important safety related points will be marked with the symbol.
- If the equipment is used in a manner not specified by this manual, the protection provided by this equipment may be impaired.



3. Document Management

We recommend that you keep this manual, along with the factory test report close to the cabinet for easy reference by the cabinet operator and qualified maintenance personnel.

If you require replacements for any of the provided documentation (including factory test reports) you can request copies from Esco Customer Services. * Provide the following information when making requests for replacement documents:

- Company (Organization) Name
- Product Brand and Model
- Product Serial Number
- Documents requested

* There may be a nominal charge for this service.

4. Limitation of Liability

The disposal and / or emission of substances used in connection with this equipment may be governed by various local regulations. Familiarization and compliance with any such regulations are

the sole responsibility of the users. Esco's liability is limited with respect to user compliance with such regulations.

5. European Union Directive on WEEE and RoHS

The European Union has issued two directives:

- **Directive 2002/96/EC on Waste Electrical and Electronic Equipment (WEEE)**

This product is required to comply with the European Union's Waste Electrical & Electronic Equipment (WEEE) Directive 2002/96/EC. It is marked with the following symbol:

Esco sells products through distributors throughout Europe. Contact your local Esco distributor for recycling/disposal.

Recommended method of disposal is according to The Federal, State and Local Government regulations.



- **Directive 2002/95/EC on Restriction on the use of Hazardous Substances (RoHS)**

With respect to the directive on RoHS, please note that this cabinet falls under category 8 (medical devices) and category 9 (monitoring and control instruments) and is therefore exempted from requirement to comply with the provisions of this directive.

6. Symbols

Information in this manual may be prefaced with the following symbols. They are provided to help you identify important operational, safety, maintenance or conformance issues.



Electrical Hazard: Danger of electric shock



Turn off and disconnect from main supply before proceeding: Do not perform this operation while the unit is operational



The Biohazard Symbol on the front panel of the cabinet indicates the presence of biological substances that pose a threat to human health

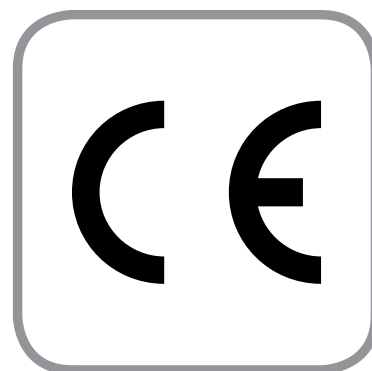


Approved Service Engineer Only: Operation to be performed only by approved engineers

Declaration of Conformity

In accordance to EN ISO/IEC 17050-1:2010

We, Esco Micro Pte Ltd
of 21 Changi South Street 1
Singapore, 486777
Tel: +65 6542 0833
Fax: +65 6542 6920



declare on our sole responsibility that the product:

Category : Class II Biological Safety Cabinet
Brand : Airstream
Model : AC2-2E8, AC2-3E8, AC2-4E8, AC2-5E8, AC2-6E8, AC2-3E8-M, AC2-4E8-M, AC2-5E8-M, AC2-6E8-M, AC2-2S8, AC2-3S8, AC2-4S8, AC2-5S8, AC2-6S8, AC2-3S8-M, AC2-4S8-M, AC2-5S8-M, AC2-6S8-M

in accordance with the following directives:

2014/35/EU : The Low Voltage Directive and its amending directives
2014/30/EU : The Electromagnetic Compatibility Directive and its amending directives
2011/65/EU : The RoHS in Electrical and Electronic Equipment Directive and its amending directives

has been designed to comply with the requirement of the following Harmonized Standard:

Low Voltage : EN 61010-1:2010
EMC : EN 61326-1:2013 Class B
Design/ : EN 12469 (2000) Class I Microbiological Safety Cabinet
Performance
Criteria

More information may be obtained from Esco's authorized distributors located within the European Union. A list of these parties and their contact information is available on request from Esco.

Lim Lay Yew

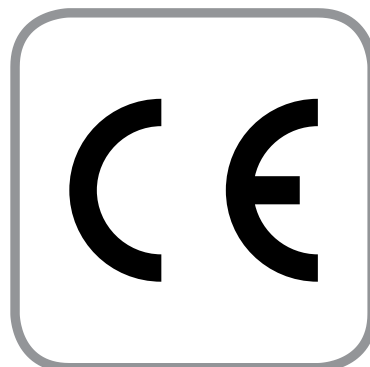
Lim Lay Yew
Director, Esco

This Declaration of Conformity is only applicable for 230VAC 50/60 Hz units

Declaration of Conformity

In accordance to EN ISO/IEC 17050-1:2010

We, Esco Micro Pte Ltd
of 21 Changi South Street 1
Singapore, 486777
Tel: +65 6542 0833
Fax: +65 6542 6920



declare on our sole responsibility that the product:

Category : Class II Biological Safety Cabinet
Brand : Airstream Duo
Model : AC2-4D8, AC2-6D8, AC2-4G8, AC2-6G8

in accordance with the following directives:

2014/35/EU : The Low Voltage Directive and its amending directives
2014/30/EU : The Electromagnetic Compatibility Directive and its amending directives
2011/65/EU : The RoHS in Electrical and Electronic Equipment Directive and its amending directives

has been designed to comply with the requirement of the following Harmonized Standard:

Low Voltage : EN 61010-1:2010
EMC : EN 61326-1:2013 Class B
Design/ : EN 12469 (2000) Class I Microbiological Safety Cabinet
Performance
Criteria

More information may be obtained from Esco's authorized distributors located within the European Union. A list of these parties and their contact information is available on request from Esco.

Lim Lay Yew

Lim Lay Yew
Director, Esco

This Declaration of Conformity is only applicable for 230VAC 50/60 Hz units

Class II Biological Safety Cabinet **ESCO**

Chapter 1 – Maintenance and Recertification of the Cabinet

Esco products generally provide years of trouble-free operation however like all equipment they require maintenance and service. Maintenance and service should be carried out by trained personnel. Esco offers training courses to equip service providers with the latest skills, information and tools to successfully maintain and service Esco products. For more information on the nearest training course, please contact Esco.

Service providers should familiarize themselves with the basic operating principles of products before working on them. Suitable references include information in this Manual or brochures which may be downloaded from the Esco web site.

Biological safety cabinets generally require:

- Re-certification, when:
 - The cabinet is re-located.
 - Cabinet performance is suspect.
 - After filter or blower replacement.
 - At least once a year or based on your risk assessment.
 - During recertification:
 - Cabinet airflow velocities and flow patterns are verified against the manufacturer's specifications and relevant international standards
 - Filters are scan-tested to ensure they do not leak.
 - Operator comfort tests may be performed.
 - If airflow velocities are found to be off setpoint, adjustments are made as part of the certification process before final values are recorded.
- Airflow alarm calibration, when:
 - The cause of the airflow alarm cannot be determined.
 - Re-certification indicates the displayed airflow deviates by > 0.02 m/s (4 fpm) from actual measured velocities (only for products with airflow velocity displays).
- Decontamination before filter or fan replacement, or as specified.
- Filter replacement, when:
 - The filters are clogged and the fan(s) are unable to compensate for the filter loading.
 - Filter leaks which cannot be repaired are found during scan-testing.

Note: Before filter replacement, the cabinet must be decontaminated.
- Fan replacement (typically rare) if failure occurs.
- Routine maintenance also includes:
 - Fluorescent lamp(s) replacement - typically once every 2 years.
 - UV lamp replacement - typically once every 1 year.

1.1. Installation Check

1. Check that the cabinet was installed away from air conditioning unit, heater, door, ceiling lamp, or any other equipment that affect cabinet airflow to ensure the sensor reading on top of cabinet is undisturbed.
2. Make sure the holes on the exhaust airflow sensor are not blocked, and the holes on the exhaust and downflow airflow sensors are perpendicular to the airflow path. If the airflow sensor is moved, then the microprocessor requires re-calibration.
3. Make sure the holes on the exhaust and downflow airflow sensors tube are perpendicular to the airflow path. If the airflow sensor tube is moved, then the microprocessor requires re-calibration.

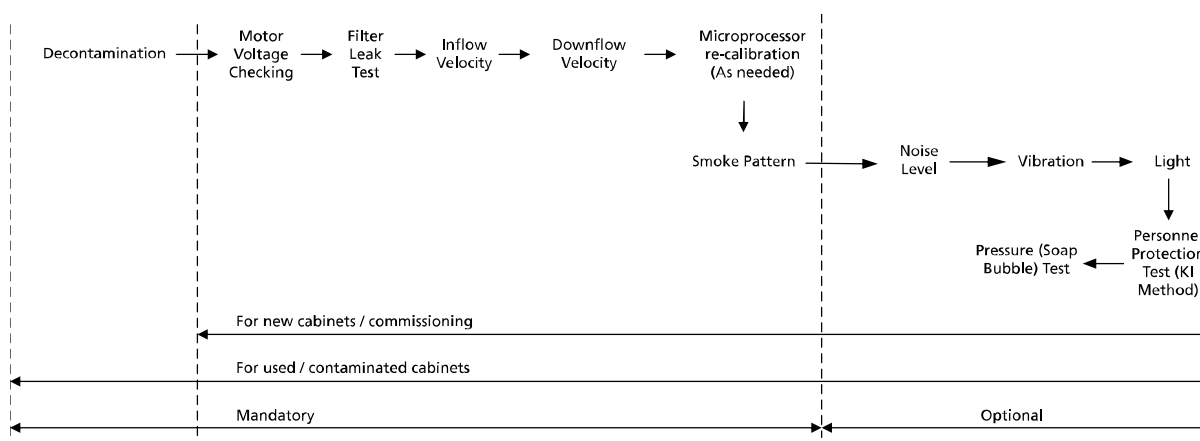
4. Make sure that the exhaust filter is not blocked, and the downflow diffuser is properly installed.
5. Make sure that the downflow diffuser is properly installed.
6. Check that the fan duty cycle matches with the value given on the test report.

1.2. Cabinet Field Certification

Cabinet field certification must be performed on:

- New installation.
- Relocation of cabinet.
- Activation of airflow alarm with undetermined cause.
- When cabinet performance is suspect.
- At least once in a year or based on your risk assessment.

The steps to perform field certification:



1.3. Performing Calibration / Certification

1.3.1. Setting Cabinet in Maintenance Mode

In Maintenance Mode all alarms are defeated, and the window can be moved without the light being switched off.

1. At normal mode LCD displays:

ESCO AC2	XX: XX
Sash: OK	AirFlow: OK
I: 0.45 m/s	D: 0.30 m/s

OR

ESCO AC2	XX: XX
Sash: OK	AirFlow: OK
I: 89 fpm	D: 59 fpm

2. Press MENU button to accessing microprocessor menu. The LCD will display:

ESCO AC2	XX: XX
PASSWORD: 0	
Use ↑ / ↓ to change	
Then press set...	

Use the ▼ button until the cursor on the LCD is at “ENTER MENU”. Press the SET button. ‘PASSWORD’ with flashing ‘0’ will be displayed. Enter the ADMIN PIN. Follow procedure below to enter the default ADMIN PIN of 0009.

- Press SET button. Flashing ‘00’ will be displayed.
 - Press SET button. Flashing ‘000’ will be displayed.
 - Press SET button. Flashing ‘0000’ will be displayed.
 - Press ▼ button to key in 9
 - Press SET button
3. Alarm will sound, wait until alarm stops. Display should read:

MENU OPTIONS	XX: XX
→SETTINGS	
SET MODE	
FIELD CALIB	

4. Use the ▼ button until the cursor on the LCD is at “SET MODE”. Press the SET button.

MENU OPTIONS	XX: XX
NORMAL MODE	
QUICKSTART MODE	
→MAINTENANCE MODE	

5. Use the ▼ button until the cursor is at “MAINTENANCE MODE”. Press the SET button. Now the cabinet is in maintenance mode. Press the MENU button twice to exit from the menu mode:
6. The word “MAINTENANCE” is displayed on the LCD as shown below:

MAINTENANCE	XX: XX
Sash: OK	
I: 89 fpm	D: 59 fpm

OR

MAINTENANCE	XX: XX
Sash: OK	
I: 0.45 m/s	D: 0.30 m/s

1.3.2. Adjusting Airflow

Carry out the steps below to adjust the airflow:

1. Remove the two screws holding the front panel (one screw located on the blue panel at each end on bottom edge).
2. Lift front panel. It will be held open by the front panel stick.
3. Locate speed controller positioned toward right side of electrical panel.
4. Plug the multimeter probes to the Motor Voltage Sampling Port.
5. Adjust the airflow by adjusting the speed controller.

1.3.2.1. Initial Voltage Adjustment (only for AC2- _D_ and AC2- _G_)

These AC2 BSCs are equipped with an exhaust fan below of the exhaust filter. This fan needs to be adjusted so that the correct balance between downflow velocity and inflow velocity is achieved.

No.	Main fan	Exhaust fan	Inflow	Downflow
1	+	+	++	+
2	+	=	+	+
3	+	-	++	++
4	=	+	-	-
5	=	=	=	=
6	=	-	+	+
7	-	+	--	--
8	-	=	-	-
9	-	-	--	-

+ : Faster/increase in value
 ++ : Increase greatly in value
 = : Constant
 - : Slower/decrease in value
 -- : Decrease greatly in value

The BSC downflow and exhaust fan speed control need to be adjusted to get nominal downflow velocity of 0.30 ± 0.025 m/s.

1.3.2.2. Damper Adjustment

The damper is used to balance the downflow velocity and exhaust velocity. The damper needs to be adjusted when the downflow velocity does not meet the accepted criteria (refer to the cabinet's technical specifications). Open the damper opening if the downflow velocity is too high; close the damper opening if the downflow velocity is too low.

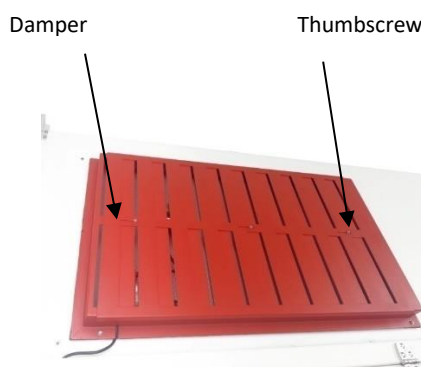


Figure 1.1. Adjustment of Damper

Loosen the thumbscrew(s) to slide the damper left or right to increase or reduce the downflow velocity.

1.4. Parts Replacement

The use of non-Esco parts and / or parts not supplied directly by Esco or our authorized distributors, including but not limited to maintenance parts, spare parts, replacement parts, system components and / or system accessories, shall void all expressed or implied warranties.

1.4.1. Filter Replacement Procedure

The filters should not be replaced until the entire cabinet has been decontaminated.

Under normal usage and barring any accident (example - puncture), the two filters do not need replacement unless and until the inflow and / or downflow velocity can no longer be maintained within the specifications.

1. Before the filters are removed from the BSC, their contaminated faces should be taped off using plastic, cardboard or some other suitable material. This will minimize the risk of personnel in the event the decontamination was not thorough.
2. Used filters should be disposed in accordance with local regulations. They may have to be incinerated as medical waste. They should be double bagged and appropriately labeled after being removed from the cabinet.
3. Proper PPE (based on your risk assessment) should be worn while removing used filters. You should dispose of these things after the procedure. Finally you should properly wash your hands.
4. Before the new filters are installed, all surfaces should be thoroughly cleaned of silicon and/or adherent gasket material. The new filter should be carefully handled and examined before being fitted. It is important that the filters and the gaskets be checked for leaks prior to use.

Two persons may be required to facilitate filter changing on 5' and 6' BSC.

Two steps of tightening are necessary where the first round torque strength should be about one-third of the final torque required. Over tightening may cause filter damage and potential leak.

1.4.1.1. Exhaust Filter Replacement for AC2-_E_ and AC2-_S_ Class II BSC

1. Move the Sash Window down until it reaches the armrest.
Note: For motorized sash BSC, remove the window straps after moving the sash window to the lowest position.
2. Remove red plenum cover using Phillips screwdriver.
3. Unclip those four red clips to lower down the sliding filter bracket.
Caution: Be careful when lowering down the sliding bracket because it might damage the filter gasket
4. Take out the exhaust filter carefully.
5. To install new filter, just do a reverse procedure from Step 4 to Step 1.

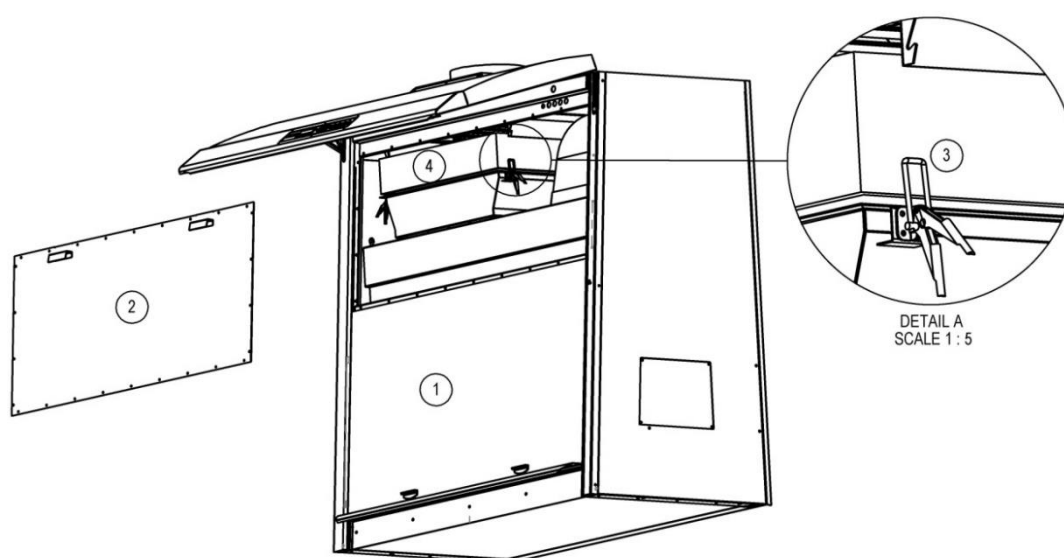


Figure 1.2. Filter Replacement for AC2-_E_ and AC2-_S_

1.4.1.2. Exhaust Filter Replacement for AC2-_D_ and AC2-_G_ Class II BSC

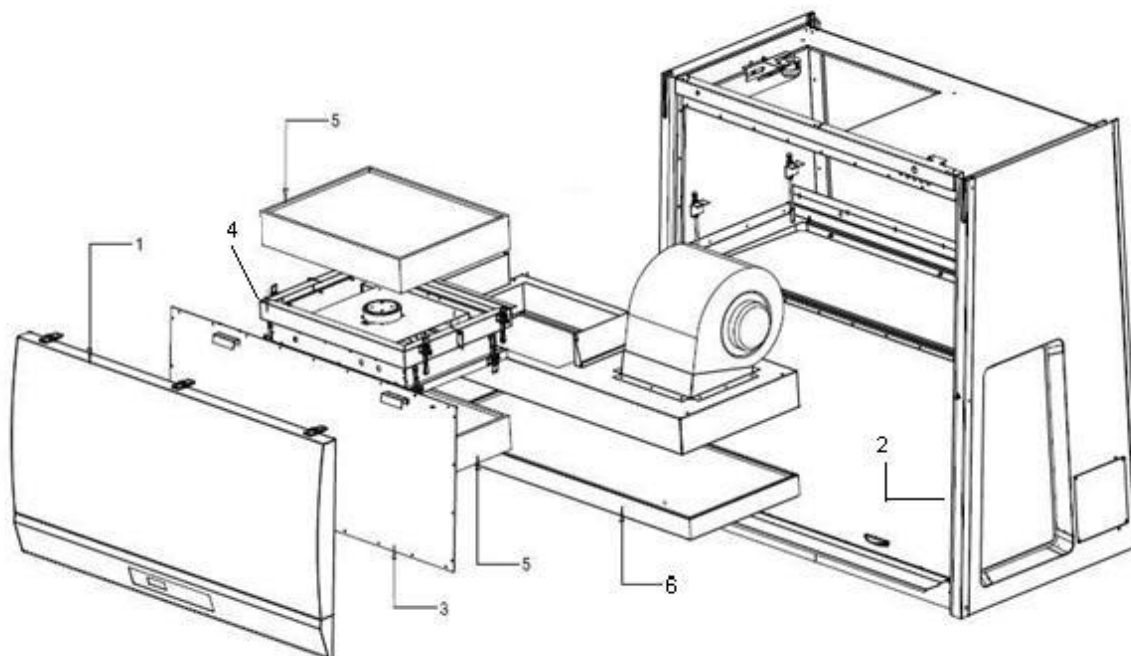


Figure 1.3. Filter Replacement for AC2-_D_ and AC2-_G_

1. Remove the screws from the bottom of blue panel and then lift up the front panel.
Caution: Use the cylindrical struts to carefully hold open the front panel. Continue to work only after the front panel is secured.
2. Move sash window to the lowest position.
Note: For motorized sash BSC, remove the window straps after moving the sash window to the lowest position.
3. Remove red plenum cover using Phillips screwdriver.
4. Remove the secondary exhaust filter mounting frame from the front.
5. Remove the secondary exhaust filter.
6. Remove the primary exhaust filter from the front.
7. To install the new filter, reverse Steps 1 to 6.
 - a. When blower housing onto new filter is secured, please ensure gasket compression is equal on both downflow and exhaust filter.
 - b. When the exhaust filter mounting frame is secured, please do not over-tighten the nuts, use the lowest torque if using a power tool.

1.4.1.3. Downflow Filter Replacement

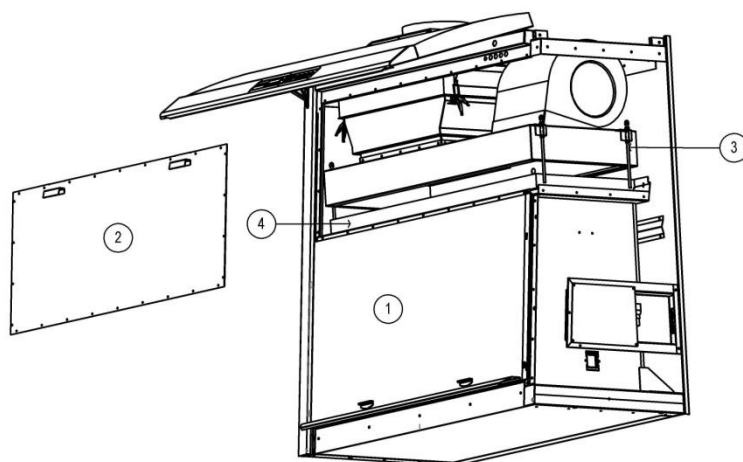


Figure 1.4. Downflow Filter Replacement

1. Move the Sash Window down until it reaches the armrest.
Note: For motorized sash BSC, remove the window straps after moving the sash window to the lowest position.
2. Remove red plenum cover using Phillips screwdriver.
3. Unclip those four red clips to lower down the sliding filter bracket.
Caution: Be careful when lowering down the sliding bracket because it might damage the filter gasket
4. Take out the downflow filter carefully.
5. To install new filter, just do a reverse procedure from Step 4 to Step 1.

1.4.2. Blower Replacement Procedure

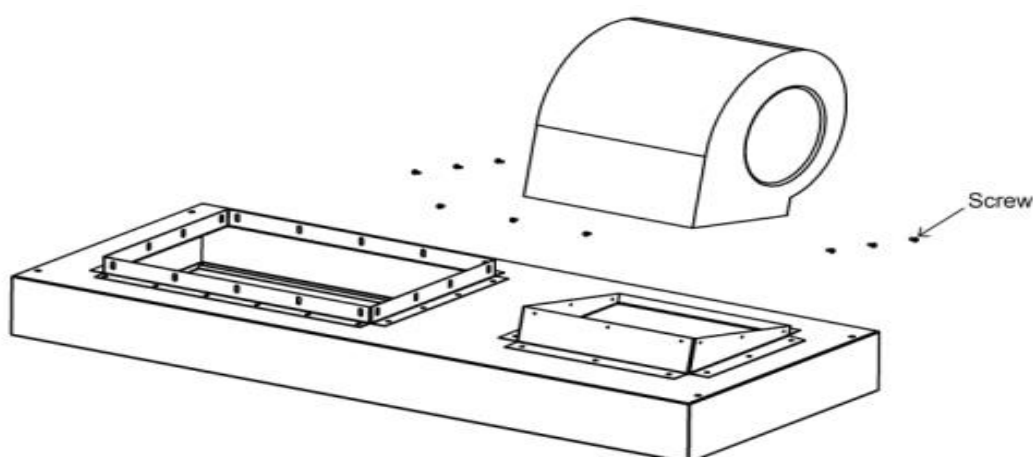


Figure 1.5. Blower Replacement

To replace the blower, do the filter replacement steps until the blower plenum assembly can be removed from the cabinet, then do the following steps:

1. Unscrew all those screws on the front and sides as shown in the image above.
2. Take out the blower carefully.

1.4.3. Fluorescent Lamp(s) Replacement Procedure

1. Disconnect the electrical connections.
2. Remove the screws from the bottom of blue panel and then lift up the front panel.
Caution: Use the cylindrical struts to carefully hold open the front panel. Continue to work only after the front panel is secured.
3. The fluorescent lamp(s) is located behind the blue panel. Disconnect the old one(s) and replace with the new one(s).

1.4.4. UV Lamp Replacement Procedure

1. Carry out necessary surface decontamination in the work zone prior to working on the UV lamp.
2. Disconnect the electrical connections.
3. Remove the old UV lamp by rotating it 90° counterclockwise and pull downward.
4. Install the new UV lamp by pushing it upward and rotating it 90° clockwise to secure the lamp in its socket.

1.4.5. Airflow Sensor Replacement Procedure

1.4.5.1. Airflow Sensor Replacement Procedure where Doughnut Sensor Housing is Used

Similar to the sensor box case, in AC2-_E_, AC2-_S_, AC2-_D_ and AC2-_G_, the airflow sensor is located on top of the cabinet. The sensor can easily be accessed in these cabinets.

1. Start by removing the damper (unscrew) on top of the cabinet (for AC2-_E_ & AC2-_S_; damper not applicable for AC2-_D_ and AC2-_G_).



Figure 1.6. Unscrewing the Damper

2. Lift the damper to locate the airflow sensor.



Figure 1.7. Lifting and Removing the Damper

3. Disconnect the airflow sensor cable connector.



Figure 1.8. Disconnecting Airflow Sensor Cable

4. Remove temperature sensor from the bracket.



Figure 1.9. Remove Temperature Sensor

5. Remove the doughnut sensor housing from the bracket.



Figure 1.10. Remove Doughnut Sensor Housi

6. Install the new sensor (it comes with the doughnut sensor housing) into the bracket and connect the sensor cable connector. Ensure the airflow direction described in the sensor housing is complied with during installation.



Figure 1.11. Airflow Sensor Direction

Chapter 2 – Decontamination

BSC decontamination should be performed in any of the following eventualities:

- Before replacing the exhaust/downflow filters or blower.
- Before accessing the contaminated negative or positive plenum of the BSC.
- In case of an accidental spillage that might have contaminated any of the inaccessible surfaces.
- Before performance validation and re-certification.
- Before BSC relocation.
- Before any procedures mandated by user's risk assessment.

Appropriate Personal Protective Equipment (PPE) per user's risk assessment that may include gloves, gown, and respirator with filter suitable for the decontamination gas/vapor must also be used by the certifier or any other qualified personnel performing the decontamination.

Esco can provide the formalin vaporizer (FV-001) and decontamination bag needed for the decontamination procedure.

The information provided in this section does not construe a final advice on how decontamination must be carried out. Appropriate personnel including the safety personnel in the facility must be consulted and adequate risk assessment must be conducted to ensure the safety and efficacy of the decontamination procedures.

2.1. Decontamination Agents

2.1.1. Formalin/Paraformaldehyde Decontamination

Typically the decontamination is performed using formalin gas by either vaporizing 37% formalin solution or by de-polymerization of solid paraformaldehyde.

Despite its widespread usage for decontamination, formalin gas presents the following health risks:

- External contact can cause irritation to skin, eyes, and mucous membranes.
- Inhalation in small concentrations can cause coughing, nausea, and diarrhea.
- Inhalation in large concentrations can cause convulsions, coma, and death.
- Long term exposure can cause cancer.

Although the Permissible Exposure Level (PEL) for formalin is 0.75 ppm, many scientists believe that there is no safe level of carcinogen exposure to humans. Therefore, typically the room must be evacuated when the decontamination process is performed, which leads to lab down time.

The use of formalin decontamination also has other disadvantages:

- The process is time-consuming.
- The certifier needs to pulse the BSC fan to circulate the formalin vapor. This can dislodge the tape holding the plastic sheet covering the exhaust filter.

- Due to excessive residue extensive cleaning must be done after decontamination and before use.

According to OSHA [Occupational Safety and Health Administration (USA)], formaldehyde Short Term Exposure Level (STEL) is 2 ppm for 15 minutes exposure, 4 times a day, minimum of 60 minutes in between exposures. Any additional local safety regulations should also be observed. Personnel should be given adequate training. The following links provide general guidelines on formaldehyde safety:

- Regulations (Standards - 29 CFR) Formaldehyde - 1910.1048, Occupational Safety and Health Standards, OSHA (Occupational Safety and Health Administration), U.S. Department of Labor: <http://www.osha.gov> 11
- OSHA Formaldehyde Fact sheet (PDF format). Available at: http://www.osha.gov/OshDoc/data_General_Facts/formaldehydefactsheet.pdf

The time required to carry out formalin decontamination is outlined below:

No.	Process	Time
1	Set-up & sealing the BSC to make it air tight	1 hour
2	Formalin vaporization	½ hour
3	Formalin contact time to obtain target log of 4-6 kill	8 – 10 hours
4	Ammonia vaporization to neutralize formalin	½ hour
5	Ammonia contact time to neutralize formalin	2 hours
6	Exhausting the ammonia residue	1 hour
7	Tear-down & cleaning the (substantial) residue	1 hour
TOTAL without ammonia neutralization		10½ – 12½ hours
TOTAL with ammonia neutralization		14 – 17 hours

Ammonia is used to neutralize formaldehyde. OSHA prescribes the ammonia STEL is 35 ppm for 15 minutes exposure, 4 times a day, minimum of 60 minutes in between exposure.

Due to the adverse health effect of formalin gas, its use has been banned in some countries such as Germany, Austria, and Switzerland. Other European countries are expected to follow suit. Two primary candidates to replace formalin decontamination are chlorine dioxide gas and hydrogen peroxide vapor.

2.1.2. Chlorine Dioxide Decontamination

Chlorine dioxide decontamination is performed by injecting chlorine gas (Cl_2) into a cylinder filled with solid sodium chlorite (NaClO_2), which generates the greenish-yellow chloride dioxide gas (ClO_2). Chlorine dioxide decontamination is much faster than formalin. Being a true gas, it spreads quickly, without the need of pulsing the BSC's blower. It can rapidly kill the micro-organisms with high efficacy with just 1 hour contact time. User must note that the concentration of chlorine gas inside the cabinet must decrease to a safe level (e.g. by using scrubber system) before user should open the air tight seal. The Short-Term Permissible Exposure Limit (STEL) for Chlorine Dioxide gas

is 0.3 ppm. The time required for the entire process of chlorine dioxide decontamination is as follows:

No.	Process	Time
1	Set-up & sealing the BSC to make it air tight	1 hour
2	Chlorine dioxide gassing	½ hour
3	Chlorine dioxide contact time	1 hour
4	Chlorine dioxide “scrubbing”	½ hour
5	Tear-down & cleaning the (minimal) residue	½ hour
TOTAL		3½ hours

Chlorine dioxide has the PEL of 0.1 ppm, compared to 0.75 ppm for formalin. In both processes airtight BSC sealing is required to protect personnel from the gas exposure.

2.1.2.1 Recommended Decontamination Sealing Method

This section gives an example of how the cabinet can be sealed when FV-011 formalin vaporizer is used. However, similar principle can be applied for other gaseous decontamination method.



1. Put the decontamination bag to enclose the entire BSC and support stand.



2. Use the 50 mm wide aluminum tape to seal the gap between the decontamination bag and the floor. To compensate for areas where the decontamination bag must turn, use multi section tape. Ensure that the two layers of tape overlap at the end points to prevent leakage.



3. Overlap the tape at the joints.



4. These 2 cables (for BSC and for vaporizer) should be combined so that only 1 cable should come out from the

decontamination bag, to minimize chance of formalin leakage.



5. To seal the cable, first make an Ω-shape aluminum tape around the cable and make a 5 cm flat portion going to both directions. Then, paste the aluminum tape to the floor.



6. Put the bottom part of the decontamination bag on top of the aluminum tape, and then seal it with another layer of aluminum tape. Use multiple layer of aluminum tape (*at least 2 layers*) on both directions to ensure leak tightness.



7. The BSC is ready for formalin decontamination.

2.1.3. Hydrogen Peroxide Decontamination

Hydrogen peroxide (H_2O_2) decontamination is performed by flash vaporization of an aqueous peroxide mixture, creating a vapor that is distributed throughout the inside the BSC.

STERIS and BIOQUELL are two major vendors of hydrogen peroxide generators. There are significant differences in operating principles.

- The STERIS principle is to avoid condensation on surfaces to minimize corrosion and optimize vapor distribution. The relative humidity inside the BSC must be lowered to 30% so that the remaining 70% relative humidity can be occupied by the hydrogen peroxide vapor.
- The BIOQUELL principle is to seek micro-condensation to achieve the kill. The generator releases tiny high-speed droplets inside the BSC.

Hydrogen peroxide vapor is non-carcinogenic, but highly effective against micro-organisms. Hydrogen peroxide (H_2O_2) vapor breaks down under catalytic action to become air and water, making it environmentally friendly and it leaves no residues. The BSC is often aerated by ducting to speed up the breaking down of hydrogen peroxide vapor.

The time needed for the entire process is outlined below:

No.	Process	Time	
1	Set-up & sealing the BSC	½ hour	
2	Conditioning and decontamination cycle	½ - 1 ½ hour	
3	Ducting out H_2O_2	H_2O_2 generator doing aeration	½ hour 8 hours
4	Tear-down	½ hour	
TOTAL		2 - 3 hrs	9½ - 10½ hrs

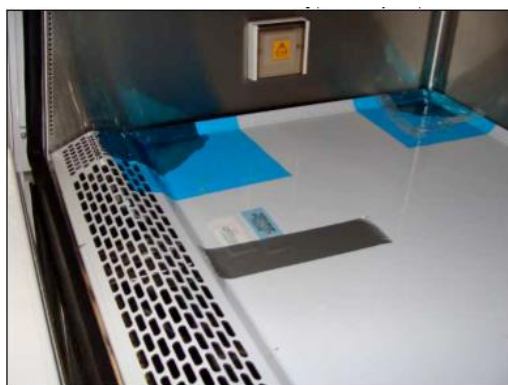
For hydrogen peroxide decontamination, the BSC need to be equipped with two ports:

1. One port located in front opening or side wall, penetrating the work zone area
2. One port located on top of the exhaust filter.

The generator used defines the port function as described below:

	Steris	Bioquell
Hydrogen peroxide source	Injected into the BSC	Generated inside the BSC
Bottom front / side port	Hydrogen peroxide introduction	Hydrogen peroxide re-introduction
Top port	Hydrogen peroxide extraction	Hydrogen peroxide extraction

The Steps to perform the hydrogen peroxide decontamination with Bioquell:



1. Place the BIs on the work tray, drain pan, exhaust:



2. Place the internet catalyst and vaporizer on tray



3. Insert the hydrogen peroxide bottle:



4. Point the vaporizer fan towards front grille:



5. Cover exhaust filter with plate with 3" exhaust port:



6. Tape exhaust plate to cabinet. Connect 3" hose:



7. Connect 3" hose to 3" inlet on external catalyst:



8. Connect 1" hose to 1" outlet on external catalyst:



9. Open the small panel on side dress panel. Connect the 1" hose to port inside side dress panel:



10. Install front Perspex. Make control cable and pressure tube come out from holes. Seal with tape.



11. Check the H_2O_2 concentration = 0 along tapes



12. After gassing period ends, mute the alarm:



13. During aeration, set 3" outlet valve to full open, to increase the airflow and speed up the process:



14. During aeration, remove tapes at perspex holes to speed up aeration. Note that internal catalyst is on:



15. Check that H_2O_2 concentration = 0 inside cabinet by inserting the probe through the hole:



16. Remove Perspex, hoses, etc. Put BIs in soy broth and incubate for 48 hours at 37°C:



17. Put a wedge on left & right sash track cover:



18. Put the wedge behind the blue panel.



19. Adjust the valve at 3" outlet to get – 10 Pa:



20. Ensure the pressure in cabinet is around – 10 Pa:



21. Press 155 ml on controller despite we use 100 ml



22. The timer on control panel starts:



23. This is how the entire set-up looks like:



24. You may see condensation that then disappears:

Chapter 3 – Troubleshooting

3.1. Electrical and Mechanical Troubleshooting

This guide addresses the most common service issues. Unless there is a note to limit the applicability of the points raised, the point applies to all AC2 series. For more troubleshooting or service information contact your local Esco Distributor.

Hardware:

- DVM (Digital Voltage Meter).
Note: An analog meter can be used for troubleshooting, but cannot be used for motor voltage measurement.
- Phillips screwdriver.
- Insulated jumper cables.

The Electrical Panel and Component Layout:

Open the front panel, the electrical panel is located on the back of the panel. A red cover is used to cover the electrical panel located on the back of the front panel of AC2-xNy model. The component layout is as followed (please note that the layout is the same regardless of the power rating).

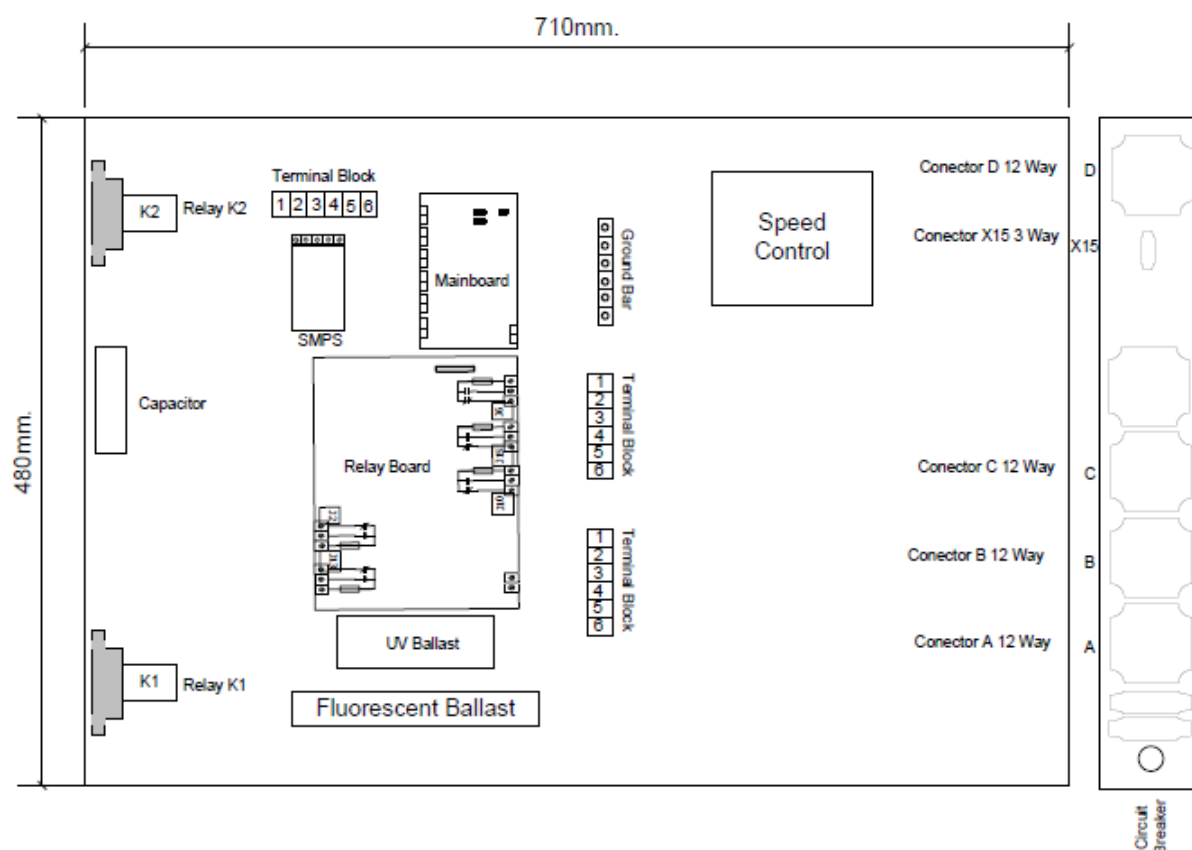


Figure 3.1. Electrical Box Layout in AC2-E/S/D/G

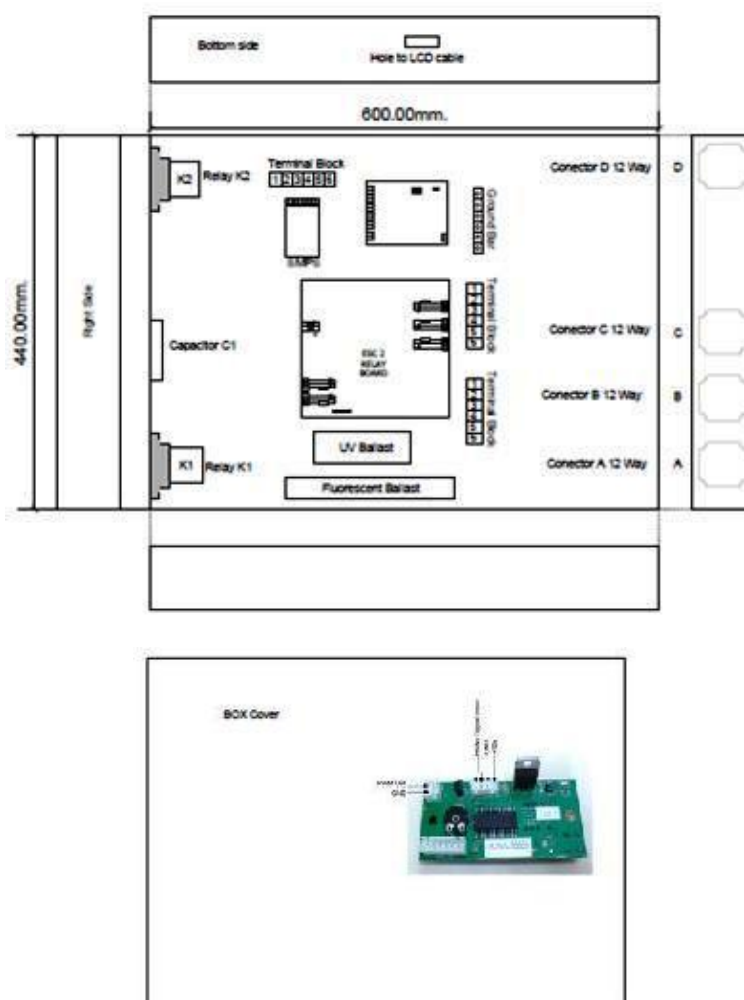


Figure 3.2. A – Electrical Box Layout in AC2 E/S/D/G; B – Electrical Box Layout in AC2-N

Problem 1: Cabinet does not start (LCD, button, fan light, and socket are inoperative)

Cause	Correction Action
Power Failure	Check wall outlet for power.
Power cord is not connected properly or faulty	- If the outlet has a switch make sure it is turned ON. - Ensure the power cord is connected correctly to the wall outlet. NOTE: Some cabinets have 2 cordsets. - Disconnect the power cord at the cabinet and measure the AC voltage between the live and the neutral terminal of the cord. - If the voltage is not within $\pm 2\%$ of the wall socket voltage, replace the cordset; otherwise Proceed to the next step.

Cause	Correction Action
Circuit breaker has tripped	- Check the circuit breaker inside the electrical panel. NOTE: If circuit breaker has tripped, do not reset the breaker before checking all electrical components and wiring connections. (See Figure 1-1 in the next page). - Does the cabinet operate correctly after resetting the circuit breaker? If it does not Proceed to the next step.

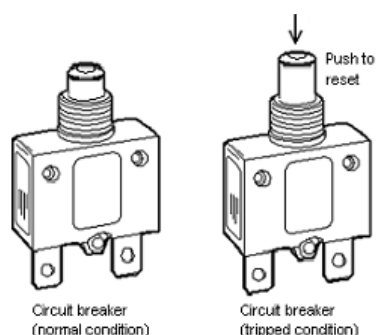


Figure 1.1

Improper connection

- Measure AC voltage between LIVE (RED) and NEUTRAL (BLUE) terminal blocks inside electrical box. See Figure 1-2.
- Voltage must be within $\pm 10\%$ of the rated voltage.
- Refer to cabinet Serial Number Tag for required voltage.
- If the voltage is out of the range or there is no voltage, check cable connection on both sides of connector A. See Figure 1-3.
 - 230/115 VAC at Connector A pin 1 and 2. Pin 3 is ground.
- If there is voltage at Connector A but no voltage is measured at the RED (LIVE) and BLUE (NEUTRAL) terminal blocks, the problem is between connector A and the terminal strips.
- If voltage is not present at connector A, but voltage is measured at the female connector for the line cord at the top of the cabinet that the power cord is plugged into, the problem is between the female connector and the pins on Connector A.

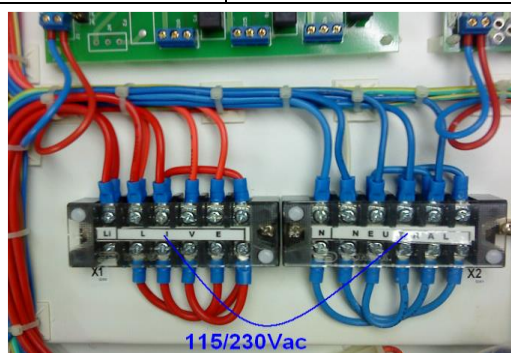


Figure 1.2

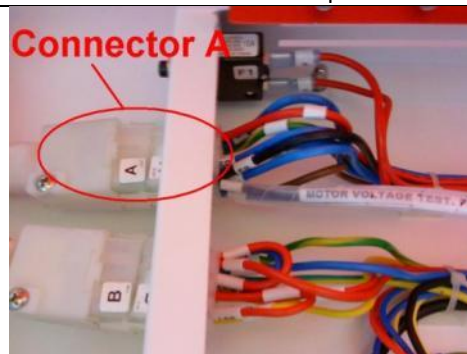
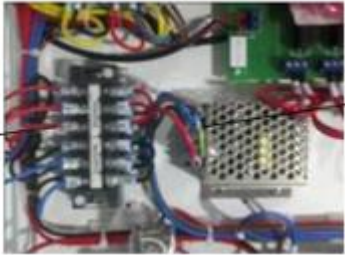
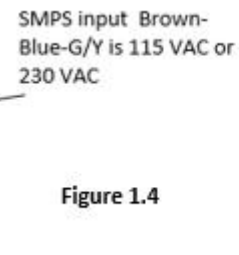
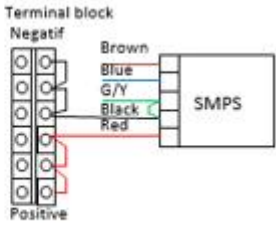


Figure 1.3

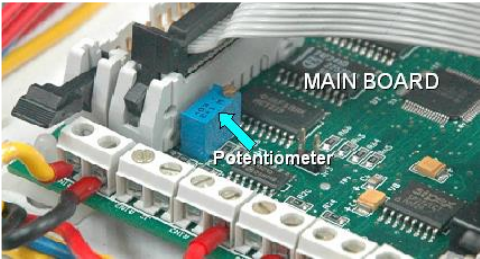
Defective power supply – Switch Mode Power supply (SMPS) (Only when SMPS item code

- Locate the SMPS (Power Supply) and Terminal block.
- On the side of terminal block, measure the DC voltage between Red (pin 3) and Black (pin 4) Measure on the side away from the SMPS.
- The voltage should be in range of $+12\text{ VDC} \pm 10\%$.
- If you do not have voltage, measures the other side of terminal block.

Cause	Correction Action
1080945) is installed)	• If out of range or no voltage, check incoming power to the SMPS – check terminal where the cord is connected. The input of SMPS should be $\pm 10\%$ of the cabinet incoming voltage. • If input voltage exists but there is no output voltage, please replace the SMPS. • If input and output voltage is correct, please Proceed to the next step.

 SMPS output Red- Black/blue is 12 VDC	 SMPS input Brown- Blue-G/Y is 115 VAC or 230 VAC Figure 1.4  Terminal block Negatif Positive Brown Blue G/Y Black Red SMPS
Connection problem to main board or from main board to display	• Measure the incoming voltage on the main board at terminal J 13 (Note polarity, blue cable closest to edge is negative -). See Figure 1-5. • Voltage should be between 6.75 – 8.25VDC, if 7.5 VDC SMPS (item code 1080328) is used. • Voltage should be between 10.8 – 13.2VDC, if 12 VDC SMPS (item code 1080945) is used. • If voltage is out of range or no voltage, ensure adequate connection between SMPS and main board. • Ensure adequate connection between flat ribbon cable from main board to LCD/keypad membrane. NOTE: There are two ribbon cables from the main board and the display and key pad. The larger flat cable is for the Display. The smaller flat cable is for the LEDs/keypad membrane. • If the LCD Display is on, but not the LEDs in keypad membrane, check connection on the smaller Flat Ribbon cable. • If the incoming voltage at J13 is of right voltage and flat ribbon cable are connected adequately but the LEDs on the keypad membrane and LCD display are still not turned on, please proceed to the next step.
	 Figure 1.5
Defective main board	• The main board is defective if the main board incoming supply is between 10.8 – 13.2VDC if 12 VDC SMPS is installed and: ▪ All LED's on the control panel are off. ▪ The LCD is blank. ▪ No buzzer sound. • If these conditions exist replace the main board, otherwise Proceed to the next step Note: when replacing main board, reconnect all wires correctly, any wrong wiring may result in damage.

Problem 2: Blank LCD

Cause	Corrective Action
Connection problem	• Ensure LCD large Flat Ribbon Cable adequately connects the main board with the display at the main board end. • Ensure that the cable has been inserted adequately into its socket on the main board. This is a Keyed connector with locking arms. The arms should point to the same direction as the cable. See Figure 2-1 below. • If the LCD Display is not on, check power to main board as detailed in Problem 1.
 Figure 2.1	
Contrast problem	• Adjust the potentiometer on main board by rotating the top metal part using flat screw driver to achieve the best LCD contrast. Rotating the part in the counterclockwise direction will increase the contrast. See Figure 2-2. • If the LCD remains blank, replace it.
 Figure 2.2	
Defective LCD	• Connect a new LCD to the LCD port on main board. • If the new LCD functions properly, this indicates the old one is defective. • If the new LCD does not work, replace the large Flat Ribbon Cable. • If the LCD still does not work replace, the main board.

Problem 3: Inoperative Buttons

Cause	Corrective Action
Connection problem	• Does the LCD display work? If it does not, please refer to Problem 1 and ensure that the main board is powered. • Ensure that the small Flat Ribbon Cable is connected to the main board – See figure 3-1. The triangle sign on the female connector of the cable indicates PIN number 1. • Interface board and membrane/keypad are located behind the blue panel, underneath the metal light deflector. • Turn off the cabinet and then remove the bulbs and the clips that hold the bulbs. See Figure 3-2. • Ensure adequate connections between the cable from the main board and the Interface board. See Figure 3-3. • Ensure adequate connections between the green plastic cable from the Interface board and the Display board. Figure 3-4.



Figure 3.1



Figure 3.2

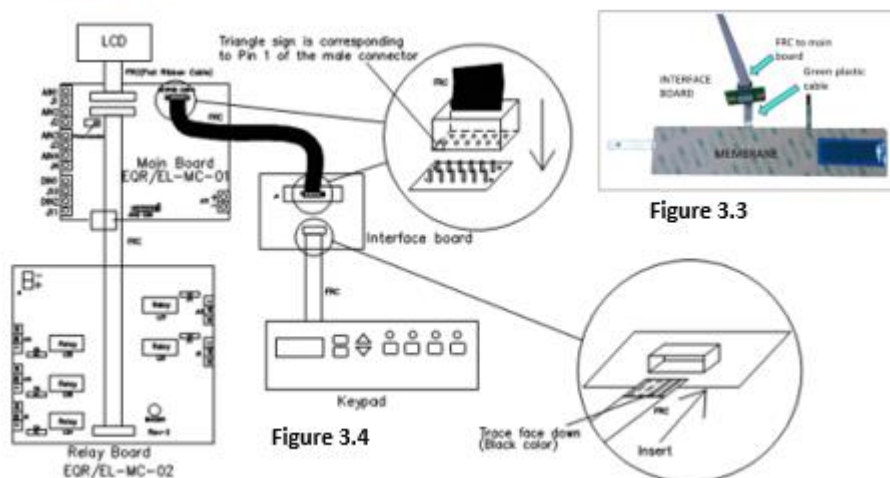


Figure 3.3

Figure 3.4

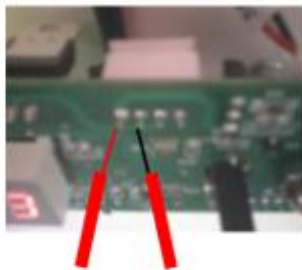
Defective Cable and / or Interface Board and / or keypad

- Replace them one by one, to check which one(s) is/are Defective.
- Replace the Defective part(s).

Problem 4: Blower Does not Function

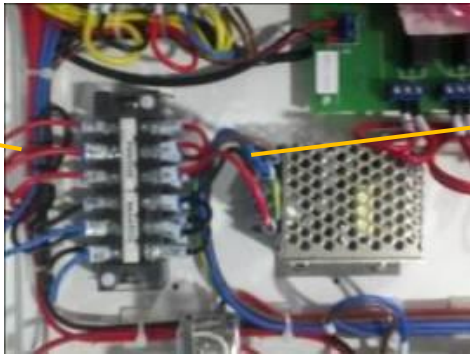
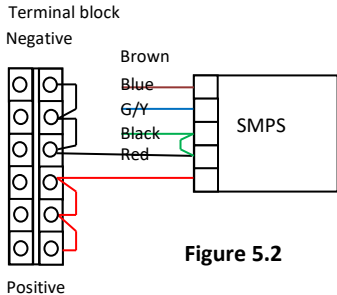
Cause	Corrective Action
Motor Blower Fan is not running	• Switch on the Fan by pressing Fan button on control panel. • Enter the Fan PIN number if required (default is 0001). • The LED for FAN should illuminate and the FAN should start. • If the FAN does not operate, Proceed to the next step. • If the LED on the cabinet front does not illuminate, please refer to Problem 1.
DC Voltage Connection problem to relay board	• Measure the incoming voltage on the Relay Board at terminal J 1 (Note polarity, BLUE cable closest to edge is negative -). See Figure 4-1. • Voltage should be between 6.75 – 8.25VDC, if 7.5 VDC SMPS (item code 1080328) is used. • Voltage should be between 10.8 – 13.2VDC, if 12 VDC SMPS (item code 1080945) is used. • If voltage is out of range or no voltage, check connection between SMPS (power supply) and relay board. • If voltage is correct, Proceed to the next step.

Cause	Corrective Action
Fuse F1 is blown or circuit is not energized	• Check that AC voltage is supplied to the Relay board. • Ensure the Motor Blower FAN button and LED is energized. • Locate LS1 circuit and measure AC voltage between large BLUE (Neutral) terminal block and the end terminal (with Red cable) on the three terminal strips (J2 on relay board). See Figure 4-1. • If there is voltage to the end terminal (Terminal with cable) check the central terminal (Normally Open Circuit) to the BLUE (Neutral) terminal block. • Should there be no voltage detected at the central terminal, the fuse could have been blown or the circuit is not energized. • Check Fuse F1 on relay board. See Figure 4-1. • If fuse F1 is blown, as temporary solution, use F5 (spare) to replace F1. • If fuse F1 is good, check the voltage between the BLUE (Neutral) terminal block and the end terminal on the LS1 circuit (J2 on relay board) • Detection of voltage at the end terminal with no cable indicates that the F1 fuse is good and the circuit is not energized.
Figure 4.1 Power into FAN Circuit Power out to ECMS Normally closed relay connection -not used	
Defective ECMS motor speed controller	• Check Voltage supply to the ECMS, check from DC supply that one pin must be 12 volt. • If there is no voltage supplied to the ECMS, check the cable connection. • If cable connection is fine, check voltage output of the DC power supply. • If there is no DC Power supply output, the DC power supply is faulty and needs replacement. • If there is voltage supplied to ECMS, proceed to the next step. • Check the display of ECMS whether or not there is any reading. • If ECMS displays any value, try to turn the potentiometer to maximum and minimum to check for response. • If ECMS responds to the adjustment, check PWM output with the multimeter. • If ECMS produces PWM output, ECMS is likely not defective. • If ECMS does not produce PWM output, ECMS is defective and needs replacement. • If ECMS does not respond after adjustment attempt with the potentiometer, ECMS is defective and needs replacement.

	 Figure 4-2 ECMS  Figure 4-3 Measure ECMS Supply  Figure 4-4 ECMS no Display  Figure 4-5 Measure PWM Output
Motor failure	<u>WARNING:</u> The cabinet MUST be decontaminated before opening the plenum cover. • Check for physical damage. • Will the motor rotate? • Check wiring. • Replace the blower.

Problem 5: Airflow Failure (AIRFAIL) – Alarm is Triggered

Cause	Corrective Action
External air interference	• Ensure that there are no external sources of airflow disturbance close to the sensor. Check vents, lights, ceiling too close, etc. • If there is no external air interference, Proceed to the next step.
Blocked cabinet grille	• Ensure that the air grilles inside the cabinet are not blocked and that the cabinet is not excessively loaded. • If there is no blockage in air grilles, Proceed to the next step.
Low building supply voltage (if new cabinet, check this first)	• Refer to the electrical layout and find motor voltage Test Point shown in Figures 4-2 and 4-3. • <u>NOTE:</u> The voltage adjustments should only be carried out by a qualified Certifier. • For a new cabinet, if the motor voltage is lower than the Nominal Voltage value recorded in the factory test report, adjust the speed controller to get output voltage to blower as specified in the Test Report – Test Conditions Documentation Sheet. • For cabinet that has been used for more than 1 year, refer to last motor Nominal Voltage from last certification report or if not available, measure the actual airflow on cabinet and increase motor voltage accordingly to achieve nominal airflow velocity. • If supply voltage is same as Nominal voltage, proceed to the next step.
Defective airflow sensor power supply – Switch Mode Power supply (SMPS) (Only when SMPS)	• Cabinet must be turned on to perform this test. • Locate the 5 pin connector. • On the BACK side of the 5 pin connector measure the DC voltage between Green (pin 5) and Black (pin 4). Measure on the side, away from the SMPS. See Figure 1-4 and 1-5. • The voltage should be in range of +12VDC $\pm$ 10%.

(item code 1080328) is installed)	• Voltage on the other side of the 5-pin connector does not need to be measured. • If out of range, check incoming power to the SMPS Molded cord – check the terminal where the cord is connected. The input should be $\pm 10\%$ of the cabinet voltage • If you have input voltage but no output voltage replace the SMPS.
Defective airflow sensor power supply – Switch Mode Power supply (Only when SMPS (item code 1080945) is installed)	• Cabinet must be turned on to perform this test. • Locate the terminal block. • On the side of the terminal block, measure the DC voltage between red and black pins. Measure on the side, away from the SMPS. See Figure 5-1 and 5-2. • The voltage should be in range of $+12\text{VDC} \pm 10\%$. • The voltage has to be measured on the other side of terminal block and SMPS. • If out of range, check incoming power to the SMPS – check terminal where the cord is connected. The input of SMPS should be $\pm 10\%$ of the cabinet incoming voltage • If input voltage is measured, but no output voltage, then replace the SMPS.
SMPS output Red – Black/blue is 12VDC  SMPS input Brown- Blue-Green Yellow is 115V AC or 230V AC Figure 5.1 Terminal block Negative  Positive Figure 5.2	
Cabinet requires the air flows to be set	• Refer to the Test Report for this model and set the air flow values to the required values. This information is also contained on the label at the bottom of the BLUE front panel.
Filter loaded or wrong speed controller setting	• Plug in the voltmeter to the Motor Voltage Sampling Port. • Measure the actual airflow velocity using anemometer or flow hood. If actual airflow is good but LCD still shows AIRFAIL, Proceed to the next step. If actual airflow is not good, please adjust motor speed controller accordingly until alarm is turned off and the LCD shows nominal airflow reading. After this, if LCD still shows AIRFAIL, Proceed to the next step. • The sensor will take about 2-3 minutes before it can register a new airflow setting. <u>Note:</u> <i>The cabinet must be certified at least annually.</i> • If the blower is already operating at maximum voltage and the airflow alarm is still triggered due to filter loaded, it's time to change the filter. • If filter is not loaded, Proceed to the next step.
Faulty/ inaccurate calibration	• Re-calibrate the microprocessor. Follow the calibration procedure in the test report. <u>Warning:</u> <i>To be carried out only by an authorized person.</i> <u>CHECKING CALIBRATION DATA</u> • After calibration is done, go to main display by pressing MENU button twice. • Press SET/Diagnostic button, will display the mode and software version that is currently being used on the cabinet. Example: DIAGNOSTIC MODE: NORMAL CP104D VERSION 1.5

	- Press DOWN button until you find the following: TEMPERATURE: Shows actual ambient temperature ADC IFF: ADC value for Fail Point Inflow – calculated using offset based on Inflow Nominal Point. ADC IFN: ADC value for Nominal Point Inflow – set during sensor calibration. ADC IFA: ADC value for Actual Inflow – showing real time sensor reading. ADC IF0: ADC value for factory calibrated Zero Point Inflow (no inflow)-set during zero calibration. ADC IF1: ADC value for factory calibrated Fail Point Inflow. ADC IF2: ADC value for factory calibrated Nominal Point Inflow. ADC DFA: ADC value for Actual Downflow – showing real time sensor reading. ADC DF0: ADC value for factory calibrated Zero Point Downflow (no downflow). ADC DFN: ADC value for Nominal Point downflow – based on field calibration. CONSTANT: value for Inflow sensor constant - keyed in during sensor calibration. CALIB TEMP: Temperature when the factory calibration was performed. ADC TEMP: ADC value for temperature. Note: Depending on preset unit at SETTINGS>MEASUREMENT UNIT> (Metric or Imperial), the units for temperature and velocity will change accordingly. For Metric- °C, m/s while for Imperial-°F, fpm. Factory setting is at Metric. - In order to prevent any AIRFAIL or unstable velocity displayed on LCD problem, please ensure the following is done after the calibration: - CONSTANT value is correct. Check it with constant written on airflow sensor body or inside Replaceable Component Record List which comes with the test report. - TEMPERATURE shows room temperature. If not, refer to next possible cause, part B. - ADC IF0 < ADC IF1 < ADC IF2. If not please re-do calibration in correct sequence: SET CONSTANT → SET ZERO → CALIB. If CALIBRATION ERROR is encountered, please refer to next possible cause, part A. - ADC DF0 < ADC DFN. If not please re-do calibration in correct sequence: SET CONSTANT → SET ZERO → CALIB. If CALIBRATION ERROR is encountered, please refer to next possible cause, part A. - Just after calibration, ADC IFA value should be close to ADC IFN value. If not, please adjust the speed controller to make both values closer by at least ± 20 units.
Sensor Failure / Sensor Misalignment	A. Air Flow Sensor failure / misalignment - Airflow sensor the top of the cabinet (E/S series model) or inside the unit below the top filter (D/G series model). Airflow sensor for D/G series models can be accessed from the back of the unit. See Figure 5-1 below. It is a stainless steel tube with 2 round holes facing up. Important to check: - Ensure that the sensor area is not blocked at all and that its through-holes are perpendicular to the filter's surface. - Exhaust sensor is attached tightly to the box / housing. - If sensor box is used, there is no gap between sensor box and filter surface. If doughnut housing is used, the installation direction is in accordance with the airflow direction. - Sensor box / housing is installed tightly to the cabinet. - Turn off the cabinet. - Open the front panel and locate electrical panel behind it.

- Refer to the Component Layout at the beginning of this section to locate the main board in the electrical box.
- Using multimeter, check the exhaust sensor's output voltage (DC) at channel AIN1(+) and J1(-) on main board (See Figure 5-2 below).
- The airflow sensor voltage should increase if airflow is increased and decrease if airflow is decreased.
- If your observations are different, check airflow sensor connection on top of the unit (see Figure 5-5) and at connector D pin 9, 10, 11, 12 on electrical panel for any bad/loose connection.
- If connections are good but sensor is still not working properly, replace it.
- If airflow sensor is working properly, Proceed to the next step.

B. Temperature Sensor Failure

- Temperature sensor is located on top of the unit, close to exhaust airflow sensor. See Figure 5-4.
- Using multimeter, check the temperature sensor's output voltage (DC) at channel AIN4(-) and J4(+) on main board See Figure 5-3.
- 0.01VDC should represent one degree Celsius. For example, if output is 0.25 VDC, it indicates temperature of 25 °C. Tolerance of $\pm 1^{\circ}\text{C}$ is acceptable.
- Replace temperature sensor if the reading is wrong.
- If temperature sensor is working properly, Proceed to the next step.



Figure 5.3

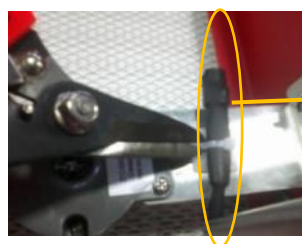


Figure 5.4



Figure 5.5

Problem 6: Excessive Blower Noise

Cause	Corrective Action
Resonance	Note: Requires a Certifier or qualified personnel • Locate the motor speed controller. • Measure the motor voltage and record this number • Increase the motor speed by 5 – 10 VAC. Does the noise change or go away? • If the noise is gone, measure the actual airflow velocity. Requires a Certifier ▪ If the velocity is still in the acceptable range, maintain it. ▪ If velocity is out of acceptable range, or if noise persists, go to next step.
Loose motor or impeller wheel mounting	Note: Decontaminate cabinet first before accessing the blower • Open the blower access panel and check if the blower mounting bolts are tight • Does the motor rotate freely without noise? • If motor is physically damaged, replace it.

Problem 7: Light Always OFF

Cause	Corrective Action
Sash in SASH ALARM state	• Is the Light LED on the front panel ON? • If the light does not turn on, press the Light button and make sure the sash is in the Normal operating position (Ready Position Normal Operating height). Note: The sash must be at the Normal Operating position. • Is the cabinet in Warm Up period? Note: The fluorescent light will not light if the cabinet is in Warm Up period unless the cabinet is in Quick Start Mode. • Move the sash to Ready position (normal operating height). • Switch on the light by pressing LIGHT button on the membrane. • If the LED is ON the keypad and main board are functioning correctly. • If fluorescent light does not light with the window in the proper position Proceed to the next step • If the LIGHT LED still does not light, skip to Ready magnetic switch Issues in this table.
Burned-out fluorescent tube	• Replace the faulty fluorescent tube. • Fluorescent tubes are located inside the blue panel. • The white lamp holder can be pulled off from the tubes.
Defective Fluorescent ballast or Relay Board issues	• Check incoming voltage to the Relay board. • Locate LS7 circuit and measure AC voltage between BLUE (NEUTRAL) terminal block and the end terminal (with Red cable) on the three-terminal strip (J13 on relay board). • If there is voltage to the end terminal (Terminal with cable), check the central terminal (Normally Open Circuit) to the BLUE (NEUTRAL) terminal block. • No voltage at the central terminal indicates a blown fuse or the circuit is not energized. • Check Fuse F7 on relay board. See Figure 7-1 below. • If fuse F7 is blown, as temporary solution, use F5 (spare) to replace F7. • If fuse F7 is good, check the voltage between the BLUE (NEUTRAL) terminal block and the end terminal on the LS7 circuit (J13 on relay board). • Voltage at the end terminal with no cable indicates the F7 fuse is good and the circuit is not energized.

Cause	Corrective Action
	- Make sure the LIGHT LED is on. - If the ballast has power and the bulbs are new the problem is either connections or the ballast. - Check Connector C connections. If there are no loose terminals, replace the ballast.

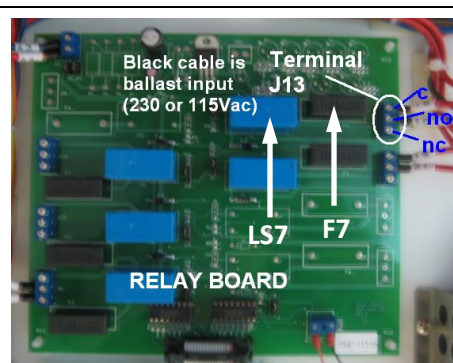


Figure 7.1

Ready magnetic switch Issues	- With the LIGHT LED ON and the sash at the Normal operating position (Ready) - Does the Display indicate "SASH: OK" or "SASH: NO!" "SASH: NO!" indicates the sash is not at the proper position and the LIGHT LED should not be on. This indicates a main board problem or a magnetic switch detection problem. - To access the magnetic switches for the sash positions, remove the right sash window profile cover as you face the cabinet. - There are 3 magnetic switches inside the right profile. See Figure 7-2 below. - The one in the middle is for Ready position. - There is a magnet attached to the glass to activate the magnet. - Shift the switch or magnet position so the distance between them is between 10-13mm (3/8 to 1/2") as explained in Figure 7-2 below. If the distance is too far, the switch may not be able to detect the magnet. - If the LCD still displays "SASH: NO!", the switch may be defective or there is a wiring connection problem. To check the connections proceed to the next step.
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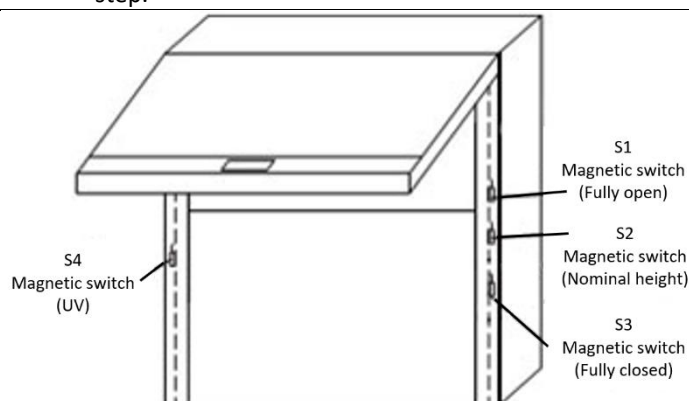


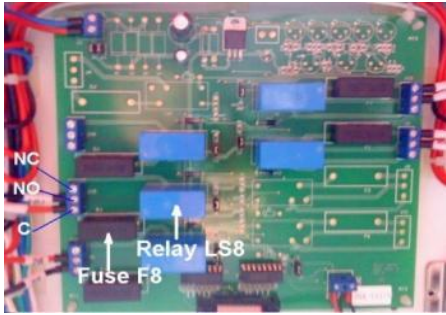
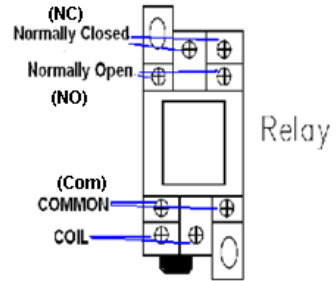
Figure 7.2

Connection Problem	- Locate Connector C and D and check male and female side for any bad connection. - If the fluorescent lamp still does not light with the LIGHT LED ON return to start of Problem 7 and repeat the troubleshooting process.
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Problem 8: UV Light Always OFF

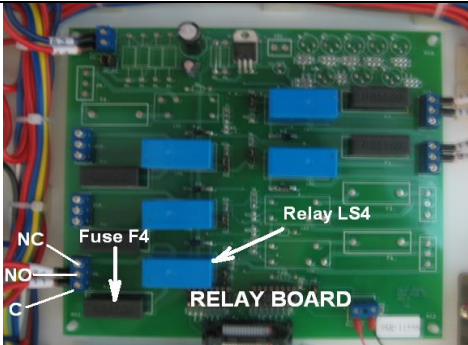
Cause	Corrective Action
Sash not in UV state	• Lower the sash to UV mode position. • If “SASH FULLY CLOSED” is not displayed on the LCD, please refer to the Magnetic Switch troubleshooting section. • Switch on the UV lamp by pressing the UV button. • If “SASH FULLY CLOSED” is displayed on LCD but UV is not ON after pressing UV button, refer to the Magnetic Switch troubleshooting section.
Faulty UV tube	• Replace the faulty UV tube.
Faulty UV ballast	• Check incoming voltage to the Relay board. • Locate LS8 (J15 Terminal) circuit and measure AC voltage between large BLUE (Neutral) terminal block and the end terminal (with Red cable) on the three terminal strip (J15 on relay board). • If there is no incoming voltage to the end terminal (Terminal with cable), refer to the next step NO power to UV Relay circuit LS8. • If voltage to the end terminal (Terminal with cable) exists, check the central terminal (Normally Open Circuit) to the BLUE (Neutral) terminal block. • Absence of voltage at the central terminal indicates a blown fuse or the circuit is not energized. • Check Fuse F8 on relay board. See Figure 8-1 below. • If fuse F8 is blown, as temporary solution, use F5 (spare) to replace F8. • If fuse F8 is good, check the voltage between the BLUE(NEUTRAL) terminal block and the end terminal on the LS8 circuit (on J15 terminal of relay board). • Voltage at the end terminal with no cable indicates the F8 fuse is good and the circuit is not energized. • Make sure the UV LED is on. • If the ballast has power and the UV lamp is new, the problem is either cable connections or the UV ballast. • Check Connector B cable connections. If there are no loose terminals, replace the UV ballast.

No Power to UV Relay circuit LS8 and K2 relay	• Absence of voltage at LS8 indicates K2 Relay is open. • Check magnetic UV interlock switch on Left side (as you face the cabinet) of sash. • Check the AC voltage to and from the K2 relay. • Absence of voltage indicates a wiring issue between the RED (Live) power in and the K2 relay. • If voltage to the K2 relay is present, but no voltage out of the K2 relay, this indicates the relay is not energized or defective. • Check the DC voltage to the coil of the K2 relay. Voltage should be 12VDC $\pm$ 10%. • No voltage indicates the magnetic switch on the Left side is not closed. Check the wiring to the switch Connector D-6 and D-3. • To simulate a closed switch, set the jumper between connector D-6 and D-3 at Connector D. • If the relay does not energize, check the wiring from the K2 coil to Connector D and the neutral wiring from K2 to the DC Power supply. • If wiring connections are OK, visually inspect K2 relay for any burnt or flash marks. • If burnt or flash marks are present, replace the relay. • If no burnt or flash marks are found, turn off the cabinet.
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	Disconnect relay cables and check for the continuity at K2 relay terminals. See figure 8-2. NC to COM has continuity while, NO to COM does not have continuity. Replace the relay if any of the terminals are found with wrong continuity configuration.
	
UV magnetic switch is defective or misaligned	- Move sash to Fully Closed position. The LCD should show SASH: FULLY CLOSED. - If the LCD display shows "SASH: NO!" the sash is not fully closed. - Check the magnetic switches. There are 3 magnetic switches behind the right sash profile cover. See Figure 7-2 above. The bottom one is for UV Mode position. - There is a magnet attached to the glass to activate the magnetic switch. - Shift the switch or magnet position so the distance between them is between 10-13mm (3/8" to 1/2") as explained in Figure 7-2 above. If the distance is too far, the switch may not be able to detect the magnet. - If the LCD still showing "SASH: NO" check the connections at Connector D pins D-4 and D-5 - To simulate a closed U/V Switch, connect the jumper between Connector D pins D-5 and D-4. - If this does not turn on the UV, return to the first step and repeat the procedures.

Problem 9: Electrical Socket Always OFF

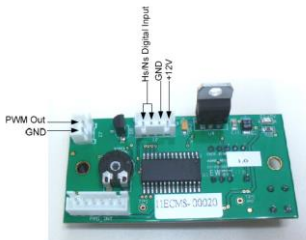
Cause	Corrective Action
Connection Problem	- Is the OUTLET LED Illuminated? If not press the Outlet button at the keypad membrane in front panel. - Check voltage coming to the Relay board. - Locate LS4 (J6 Terminal) circuit and measure AC voltage between Large BLUE (Neutral) terminal block and the end terminal (with Red cable-COM) on the three terminal strip (J6). - If there is voltage to the end terminal (with Red cable-COM), check the central terminal (Normally Open circuit) to the BLUE (Neutral) terminal block. - No voltage at the central terminal (Normally Open circuit) indicates a blown fuse or the circuit is not energized. - Check Fuse F4 on relay board. See Figure 9-1 below. - If fuse F4 is blown, replace with F5 fuse. - If fuse F4 is OK, check the voltage between the BLUE (NEUTRAL) terminal block and the end terminal (COM) on the LS4 circuit (J6 on relay board). - Voltage at the end terminal (Normally Closed circuit) with no cable indicates the F4 fuse is good and the circuit is not energized. Make sure the OUTLET LED is on. - If there is power to the relay and it is energized Check voltage across Connector B pin 1 to 2 and pin 4 to 5. - If there is power through Connector B pins, remove socket outlets from wall.

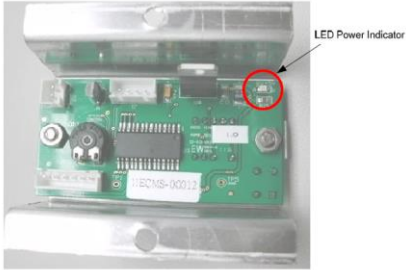
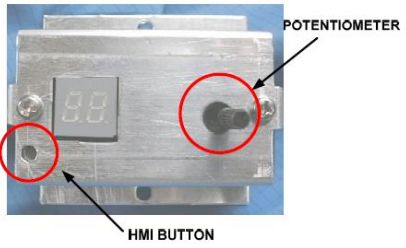
Cause	Corrective Action
	- Check incoming power to the plug in the cable to the outlet - No power indicates a wiring issue. Recheck voltage from Connector B pins. - Check for loose or faulty connection between the power cord, IEC inlet and Connector B.
	 Figure 9.1
Faulty electrical socket	- Socket(s) is(are) located inside the work zone. - Check electrical socket connection. - If connection is correct but socket has no output, replace electrical socket.

Problem 10: Contaminated Sample/Lab

Cause	Corrective Action
Blocked air grille/ Cabinet overloaded with too many items	Note: Contaminated samples are normally the result of any of the following: -Poor Technique of the user or lack of training. -Samples being contaminated in the incubator or somewhere other than the BSC. Items to check on the BSC: - Make sure that the front and back air grilles are not blocked. - Remove all equipment and other items from the cabinet and surface decontaminate the cabinet including the area under the work surface. - Surface decontaminate everything that is returned to the BSC. - If the contamination problem persists, recertify the cabinet.

Problem 11: DC Motor Speed Controller Not Functioning

Cause	Corrective Action
No output, the power indicator LEDs Off	- Turn the system on by supplying 12 VDC to the ECMS. - Replace the ECMS with a new one (Figure 11.1)  Figure 11.1 ECMS
	- Make sure that the correct DC line voltage is connected properly and turned on. - Replace the ECMS with a new one.

No output, but the power indicator LEDs on	 Figure 11.2 ECMS Showing the Power Indicator
The output voltage cannot be adjusted by turning the potentiometer	• Replace the ECMS with a new one.  Figure 11.3 ECMS Showing Potentiometer and HMI Button
The half speed mode cannot be activated	• Make sure both of the digital input has been shorted. You can check the continuity of digital input by Multimeters. • Replace the ECMS with a new one.

3.2. Software Troubleshooting

Error Code	Corrective Action
Sash: Error Position	Magnetic switches are located inside right profile cover. See drawing B-1 below.

Error Code	Corrective Action
Sash: Error Position	Magnetic switches are located inside right profile cover. See drawing B-1 below.

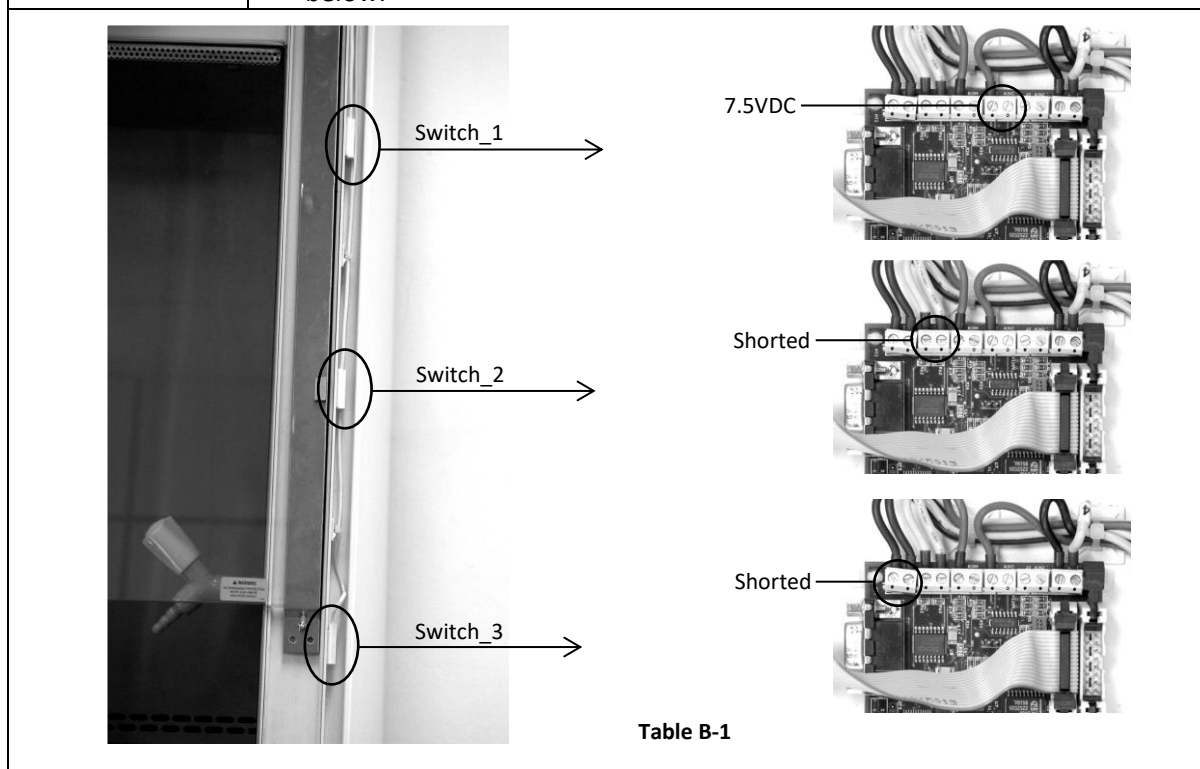


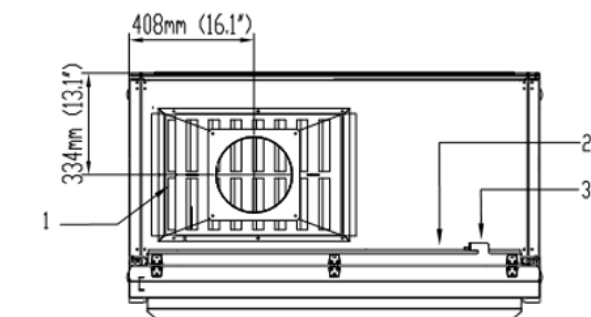
Table B-1

Figure B-2 illustrates the layout of the sash window magnetic switches. The diagram shows the sash window (front view) with four magnetic switches (Mag. Switch 1, Mag. Switch 2, Mag. Switch 3, Mag. Switch 4) and magnets. A vertical arrow indicates the sash movement range.				
Table B-2				
Magnetic Switch 1	Magnetic Switch 2	Magnetic Switch 3	Magnetic Switch 4	Description
Close	Open	Open	Open	Sash is fully opened, ALARM is given. ALARM can be muted.
Open	Close	Open	Open	Cabinet sash is at nominal height, ready to use.
Open	Open	Close	Close	Sash is fully closed; UV can be operated.

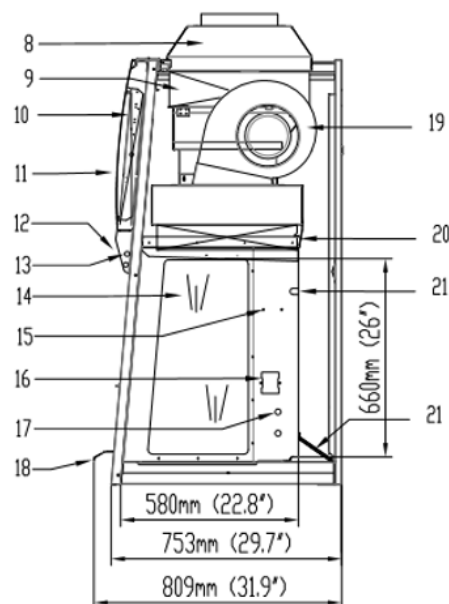
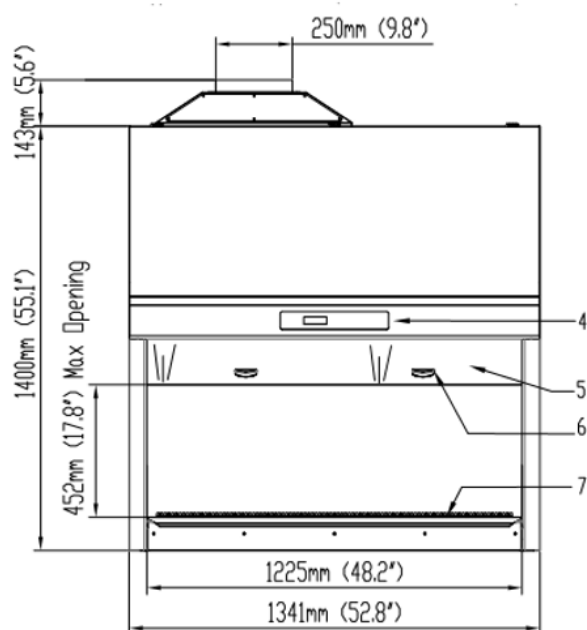
Open		Open	Open	Open	SASH is at unsafe state, ALARM is given. ALARM cannot be muted.
		Rest Conditions		SASH:ERROR POSITION detected	
SENSOR UNCALIBRATED		• Calibrate the controller. Refer to test report to calibrate.			
Sash Position	Function Available	Normal or QuickStart Mode and Air fail Alarm On	Normal or QuickStart Mode and Air fail Alarm Off	Maintenance Mode	
READY	Sash Position Detection	Yes	Yes	Yes	
	Fan Control	Yes, with WARM UP & PURGING time, FAN PIN required	Yes, with WARM UP & PURGING time, FAN PIN required	Yes, without WARM UP & PURGING time, no PIN required	
	Light Control	Yes	Yes	Yes	
	Socket Control	Yes	Yes	Yes	
	UV Control	Interlocked	Interlocked	Interlocked	
	MENU Access	Yes, USER or ADMIN PIN required	Yes, USER or ADMIN PIN required	Yes, no PIN required	
	Pressing Set button	Complete Diagnostic Message	Complete Diagnostic Message	Complete Diagnostic Message	
	Timer	Yes	Yes	No	
	Current Time Display	Yes	Yes	Yes	
	Air Velocity Display	Yes	Yes	Yes	
	If not calibrated, displaying “Sensor Uncalibrated”				
	Air Velocity Display Status	Yes	Yes	No	
Air Fail checking	Yes, Alarm and display alert if air fails	No	No		
SASH ALARM	Sash Position Detection	Yes	Yes	Yes	
	Current Time Display	Yes	Yes	Yes	
	Sash Alarm	Yes, not mutable	Yes, not mutable	Not Applicable	
	Fan Control	Yes, without WARM UP & PURGING time, no PIN required	Yes, without WARM UP & PURGING time, no PIN required	Yes, without WARM UP & PURGING time, no PIN required	
	Light Control	No	No	Yes	
	Socket Control	Yes	Yes	Yes	
	UV Control	Interlocked	Interlocked	Interlocked	
	Air Velocity Display	Yes	Yes	Yes	
	Air Velocity Display Status	Yes	Yes	No	
	Air Fail checking	Yes, display alert if air fails	No	No	
FULLY OPEN	Sash Position Detection	Yes	Yes	Yes	
	Current Time Display	Yes	Yes	Yes	
	Alarm	Yes, mutable for 5 minutes	Yes, mutable for 5 minutes	Not Applicable	
	Fan Control	Yes, with WARM UP & PURGING time, FAN PIN required	Yes, with WARM UP & PURGING time, FAN PIN required	Not Applicable	
	Light Control	Yes	Yes	Yes	
	Socket Control	Yes	Yes	Yes	
	UV CONTROL	Interlocked	Interlocked	Interlocked	
	FULLY CLOSED	Sash Position Detection	Yes	Yes	Yes
SWITCH OFF Fan requested		Yes	Yes	No	
Fan Control		Interlocked	Interlocked	Yes	
Light Control		Interlocked	Interlocked	Yes	
Socket Control		Yes	Yes	Yes	
UV Control		Yes	Yes	Interlocked	

Chapter 4 – Engineering Details

4.1. AC2-E Engineering Drawing



Cabinets Size	A	B	C	D
2'	610 mm 24"	730 mm 28.7"	334 mm 13.1"	233 mm 9.2"
3'	915 mm 36"	1035 mm 40.7"	334 mm 13.1"	331 mm 13"
4'	1220 mm 48"	1340 mm 52.7"	334 mm 13.1"	408 mm 16"
5'	1525 mm 60"	1645 mm 64.7"	334 mm 13.1"	560 mm 22"
6'	1830 mm 72"	1950 mm 76.8"	326 mm 12.8"	560 mm 22"

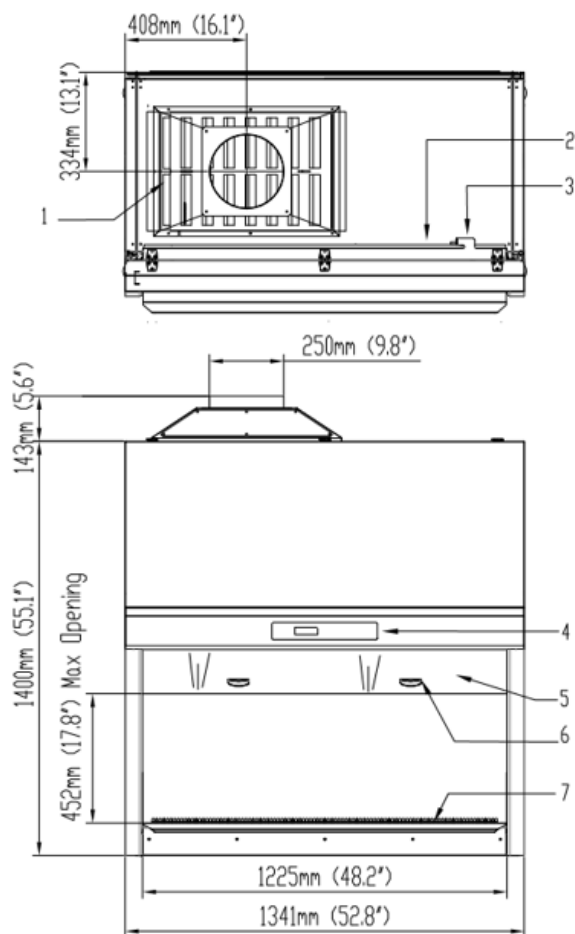


- 1 - Airflow Sensor
- 2 - RS232 Port
- 3 - Power Inlet
- 4 - Sentinel Gold Control System
- 5 - Sash Window
- 6 - Sash Handle
- 7 - Work tray

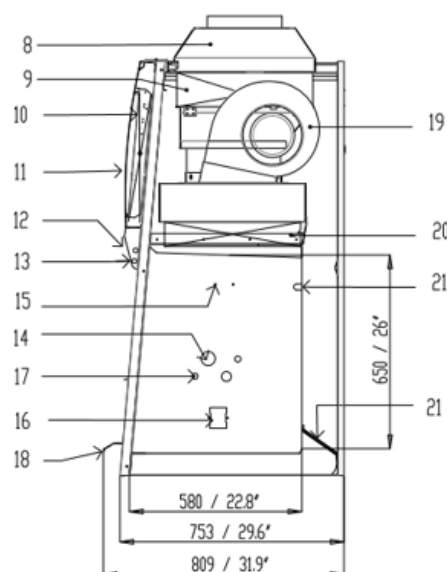
- 8 - Exhaust Collar (Optional)
- 9 - Exhaust ULPA Filter
- 10 - Electrical Panel
- 11 - Curved Front Panel
- 12 - Display Panel
- 13 - Fluorescent Lamps
- 14 - Side Window

- 15 - IV Bar Retrofit Kit Provision
- 16 - Electrical Outlet Kit Provision
- 17 - Service Fixture Kit Provision
- 18 - Stainless Steel Armrest
- 19 - ECM Blower
- 20 - Downflow ULPA Filter
- 21 - UV Lamp
- 22 - Paper Catch (optional pre-filter)

4.2. AC2-S Engineering Drawing



Cabinets Size	A	B	C	D
2'	610 mm 24"	730 mm 28.7"	334 mm 13.1"	233 mm 9.2"
3'	915 mm 36"	1035 mm 40.7"	334 mm 13.1"	331 mm 13"
4'	1220 mm 48"	1340 mm 52.7"	334 mm 13.1"	408 mm 16"
5'	1525 mm 60"	1645 mm 64.7"	334 mm 13.1"	560 mm 22"
6'	1830 mm 72"	1950 mm 76.8"	326 mm 12.8"	560 mm 22"



- 1 - Airflow Sensor
- 2 - RS232 Port
- 3 - Power Inlet
- 4 - Sentinel Gold Control System
- 5 - Sash Window
- 6 - Sash Handle
- 7 - Work tray

- 8 - Exhaust Collar (Optional)
- 9 - Exhaust ULPA Filter
- 10 - Electrical Panel
- 11 - Curved Front Panel
- 12 - Display Panel
- 13 - Fluorescent Lamps
- 14 - Steris VHP/ Bioquell HPV Port

- 15 - IV Bar Retrofit Kit Provision
- 16 - Electrical Outlet Kit Provision
- 17 - Service Fixture Kit Provision
- 18 - Stainless Steel Armrest
- 19 - ECM Blower
- 20 - Downflow ULPA Filter
- 21 - UV Lamp
- 22 - Paper Catch (optional pre-filter)

4.3. General Specifications of AC2-E and AC2-S

CLASS II TECHNICAL SPECIFICATIONS						
Glass Side: 230 V, 50/60 Hz		AC2-2E8	AC2-3E8 AC2-3E8-M	AC2-4E8 AC2-4E8-M	AC2-5E8 AC2-5E8-M	AC2-6E8 AC2-6E8-M
Stainless Steel Side : 230 V, 50/60 Hz		AC2-2S8	AC2-3S8 AC2-3S8-M	AC2-4S8 AC2-4S8-M	AC2-5S8 AC2-5S8-M	AC2-6S8 AC2-6S8-M
Nominal Size		2ft / 0.6m	3ft / 0.9m	4ft / 1.2m	5 ft / 1.5m	6 ft / 1.8m
External Dimensions * (W x D x H)	Width	730 mm (28 ¾")	1035 mm (40 ¾")	1340 mm (52 ¾")	1645 mm (64 ¾")	1950 (76 ¾")
	Depth without arm rest	753 mm (29 ½")				
	Depth with arm rest	810 mm (32")				
	Height	1400 mm (54 ¾")				
Gross Internal Dimensions (W x D x H)	Width	610 mm (24")	915 mm (36")	1220 mm (48")	1525 mm (60")	1830 mm (72")
	Depth	580 mm (22 ¾")				
	Height	660 mm (26")				
Usable Work Area		0.27 m ² (2.9 ft ²)	0.42 m ² (4.6 ft ²)	0.56 m ² (6.0 ft ²)	0.71 m ² (7.6 ft ²)	0.85 m ² (9.1 ft ²)
Tested Opening		175 mm (7")				
Working Opening		190 mm (7 ½")				
Average Airflow Velocity	Inflow	0.45 m/s (90 fpm)				
	Downflow	0.30 m/s (60 fpm)				
Airflow Volume	Inflow	173 cmh (102 cfm)	259 cmh (152 cfm)	346 cmh (204 cfm)	432 cmh (254 cfm)	519cmh (305 cfm)
	Downflow	369 cmh (217 cfm)	553 cmh (325 cfm)	738 cmh (434 cfm)	922 cmh (543 cfm)	1107 cmh (657 cfm)
	Exhaust	173 cmh (102 cfm)	259 cmh (152 cfm)	346 cmh (204 cfm)	432 cmh (254 cfm)	519cmh (305 cfm)
	Required Exhaust With Optional Thimble Exhaust Collar	260 cmh (153 cfm)	320 cmh (189 cfm)	538 cmh (317 cfm)	615 cmh (362 cfm)	823 cmh (485 cfm)
	Static Pressure For Optional Thimble Exhaust Collar	28 Pa / 0.11 in H ₂ O	29 Pa / 0.11 in H ₂ O	31 Pa / 0.12 in H ₂ O	35 Pa / 0.14 in H ₂ O	47 Pa / 0.18 in H ₂ O
ULPA Filter Efficiency		≥99.999% at 0.1 to 0.3 micron as per IEST-RP-CC001.3 USA				
		≥99.999% at MPPS as per EN 1822 (H-14) EU				
Sound Emission **	NSF / ANSI 49	56.3 dBA	56.6 dBA	58.7 dBA	58.2 dBA	59.4 dBA
	EN 12469	51.0 dBA	52.0 dBA	53.5 dBA	53.6 dBA	55.7 dBA
Fluorescent Lamp Intensity (lux)		859	1279	1404	1227	1384
Fluorescent Lamp Intensity (ft-cd)		80	119	130	114	129
Cabinet Construction	Main Body	"1.2 mm (0.05") 18-gauge electro-galvanized steel with white oven-baked epoxy-polyester isocyanate antimicrobial powder coated finish				
	Work Zone	1.5 mm (0.06") 16-gauge Stainless steel Type 304 with No. 4 finish				
	Side Walls (E Series)	UV absorbing tempered glass, 5 mm (0.2"), colourless and transparent				
	Side Walls (S Series)	1.5 mm (0.06") 16 gauge stainless steel, type 304, with 4B finish				
Electrical	Cabinet Full Load Amps (FLA)	1.8	3.5	3.7	4.3	5.5
	Heat Load (BTU/Hr)	324	447	580	717	966
Nominal Power Consumption		95	131	160	210	283
Net Weight ***		116 Kg (256 lbs)	173 Kg (381 lbs)	230 Kg (507 lbs)	288 Kg (635 lbs)	346 Kg (763 lbs)
Shipping Weight ***		143 Kg (315 lbs)	214 Kg (472 lbs)	285 Kg (628 lbs)	356 Kg (785 lbs)	428 Kg (944 lbs)
Shipping Dimensions, Maximum mm ***		850 x 820 x 1760	1120 x 820 x 1760	1450 x 820 x 1760	1720 x 820 x 1760	2050 x 820 x 1760
Shipping Volume, Maximum***		1.23 m ³	1.62 m ³	2.09 m ³	2.48 m ³	2.96 m ³

* Depth includes the remove-able arm rest and front cover. When they are removed, depth is 790 mm (31.1").

** Noise reading in open field / anechoic chamber.

*** Cabinet only, excludes optional stand.

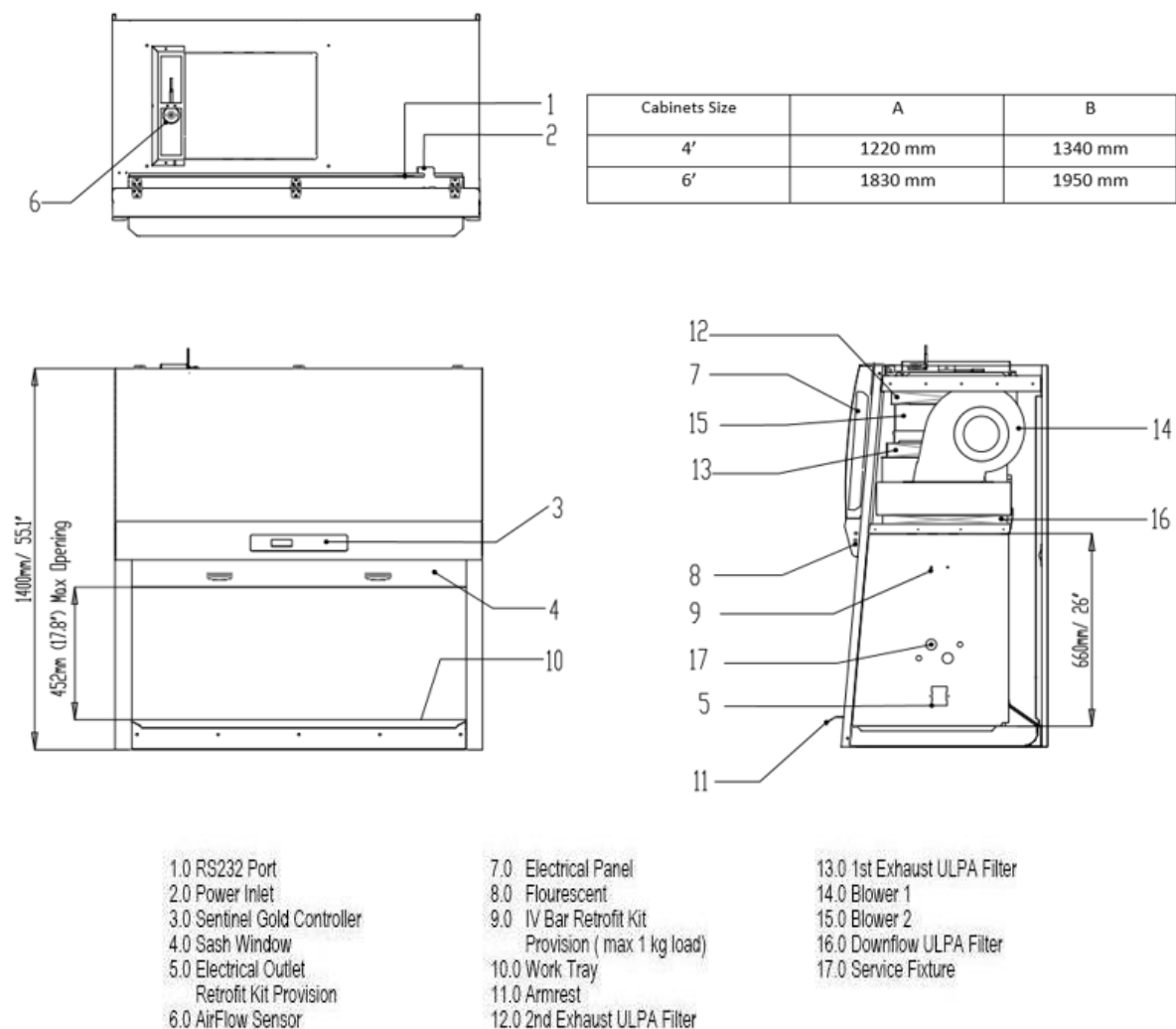
CLASS II TECHNICAL SPECIFICATIONS						
Glass Side: 115 V, 50/60 Hz		AC2-2E9	AC2-3E9 AC2-3E9-M	AC2-4E9 AC2-4E9-M	AC2-5E9 AC2-5E9-M	AC2-6E9 AC2-6E9-M
Stainless Steel Side : 115 V, 50/60 Hz		AC2-2S9	AC2-3S9 AC2-3S9-M	AC2-4S9 AC2-4S9-M	AC2-5S9 AC2-5S9-M	AC2-6S9 AC2-6S9-M
Nominal Size		2ft / 0.6m	3ft / 0.9m	4ft / 1.2m	5 ft / 1.5m	6 ft / 1.8m
External Dimensions * (W x D x H)	Width	730 mm (28 ¾")	1035 mm (40 ¾")	1340 mm (52 ¾")	1645 mm (64 ¾")	1950 (76 ¾")
	Depth without arm rest	753 mm (29 ½")				
	Depth with arm rest	810 mm (32")				
	Height	1400 mm (54 ¾")	1400 mm (54 ¾")			
Gross Internal Dimensions (W x D x H)	Width	610 mm (24")	915 mm (36")	1220 mm (48")	1525 mm (60")	1830 mm (72")
	Depth	580 mm (22 ¾")				
	Height	660 mm (26")				
Usable Work Area		0.27 m ² (2.9 ft ²)	0.42 m ² (4.6 ft ²)	0.56 m ² (6.0 ft ²)	0.71 m ² (7.6 ft ²)	0.85 m ² (9.1 ft ²)
Tested Opening		175 mm (7")				
Working Opening		190 mm (7 ½")				
Average Airflow Velocity	Inflow	0.45 m/s (90 fpm)				
	Downflow	0.30 m/s (60 fpm)				
Airflow Volume	Inflow	173 cmh (102 cfm)	259 cmh (152 cfm)	346 cmh (204 cfm)	432 cmh (254 cfm)	519cmh (305 cfm)
	Downflow	369 cmh (217 cfm)	553 cmh (325 cfm)	738 cmh (434 cfm)	922 cmh (543 cfm)	1107 cmh (657 cfm)
	Exhaust	173 cmh (102 cfm)	259 cmh (152 cfm)	346 cmh (204 cfm)	432 cmh (254 cfm)	519cmh (305 cfm)
	Required Exhaust With Optional Thimble Exhaust Collar	260 cmh (153 cfm)	320 cmh (189 cfm)	538 cmh (317 cfm)	615 cmh (362 cfm)	823 cmh (485 cfm)
	Static Pressure For Optional Thimble Exhaust Collar	28 Pa / 0.11 in H ₂ O	29 Pa / 0.11 in H ₂ O	31 Pa / 0.12 in H ₂ O	35 Pa / 0.14 in H ₂ O	47 Pa / 0.18 in H ₂ O
ULPA Filter Efficiency		≥99.999% at 0.1 to 0.3 micron as per IEST-RP-CC001.3 USA				
		≥99.999% at MPPS as per EN 1822 (H-14) EU				
Sound Emission **	NSF / ANSI 49	56.3 dBA	56.6 dBA	58.7 dBA	58.2 dBA	59.4 dBA
	EN 12469	51.0 dBA	52.0 dBA	53.5 dBA	53.6 dBA	55.7 dBA
Fluorescent Lamp Intensity (lux)		859	1279	1404	1227	1384
Fluorescent Lamp Intensity (ft-cd)		80	119	130	114	129
Cabinet Construction	Main Body	"1.2 mm (0.05") 18-gauge electro-galvanized steel with white oven-baked epoxy-polyester Isocide antimicrobial powder coated finish				
	Work Zone	1.5 mm (0.06") 16-gauge Stainless steel Type 304 with No. 4 finish				
	Side Walls (E Series)	UV absorbing tempered glass, 5 mm (0.2"), colourless and transparent				
	Side Walls (S Series)	1.5 mm (0.06") 16 gauge stainless steel, type 304, with 4B finish				
Electrical	Cabinet Full Load Amps (FLA)	1.8	3.5	3.7	4.3	5.5
	Heat Load (BTU/Hr)	324	447	580	717	966
Nominal Power Consumption		95	131	160	210	283
Net Weight ***		116 Kg (256 lbs)	173 Kg (381 lbs)	230 Kg (507 lbs)	288 Kg (635 lbs)	346 Kg (763 lbs)
Shipping Weight ***		143 Kg (315 lbs)	214 Kg (472 lbs)	285 Kg (628 lbs)	356 Kg (785 lbs)	428 Kg (944 lbs)
Shipping Dimensions, Maximum mm ***		850 x 820 x 1760	1120 x 820 x 1760	1450 x 820 x 1760	1720 x 820 x 1760	2050 x 820 x 1760
Shipping Volume, Maximum***		1.23 m ³	1.62 m ³	2.09 m ³	2.48 m ³	2.96 m ³

* Depth includes the remove-able arm rest and front cover. When they are removed, depth is 790 mm (31.1").

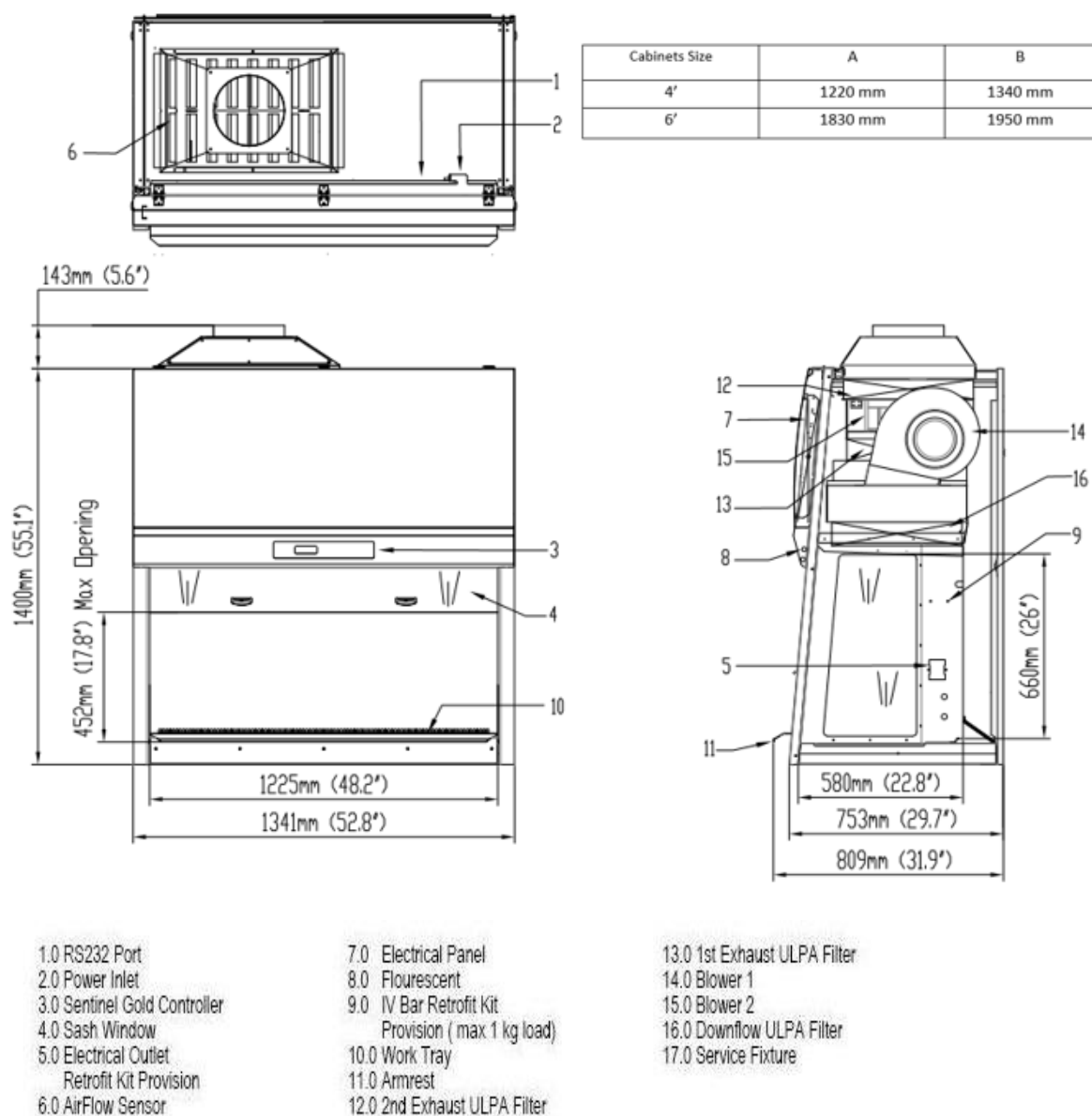
** Noise reading in open field / anechoic chamber.

*** Cabinet only, excludes optional stand.

4.4. AC2-D Engineering Drawing



4.5. AC2-G Engineering Drawing



4.6. General Specifications of AC2-D and AC2-G

AIRSTREAM CLASS II TYPE A2 TECHNICAL SPECIFICATIONS			
Glass Slide: 230 V, 50/60 Hz		AC2-4D8	AC2-6G8
Stainless Steel Side: 230 V, 50/60 Hz		AC2-4D8	AC2-6G8
Nominal Size		4 ft/ 1.2m	6 ft/ 1.8m
External Dimensions* (W x D x H)	Width	1340 mm (52 ¾")	1950 mm (76 ¾")
	Depth without arm rest	753 mm (29 ½")	
	Depth with arm rest	810 mm (32")	
	Height	1400 mm (54 ¾")	
Gross Internal Dimensions (W x D x H)	Width	1220 mm (48")	1830 mm (72")
	Depth	580 mm (22 ¾")	
	Height	660 mm (26")	
Usable Work Area		0.56 m ² (6.1 sq.ft)	0.85 m ² (9 sq.ft)
Tested Opening		175mm (7")	
Working Opening		190 mm (7 ½")	
Ave. Airflow Velocity	Inflow	0.45 m/s (90 fpm)	
	Downflow	0.30 m/s (60 fpm)	
Airflow Volume	Inflow	346 cmh (588 cfm)	519 cmh (881 cfm)
	Downflow	738 cmh (1254 cfm)	1107 cmh (1880 cfm)
	Exhaust	346 cmh (588 cfm)	519 cmh (881 cfm)
	Required Exhaust With Optional Thimble Exhaust Collar	538 m ³ / h (317 cfm)	823 m ³ / h (485 cfm)
	Static Pressure For Optional Thimble Exhaust Collar	31 Pa / 0.12 in H ₂ O	47 Pa / 0.18 in H ₂ O
ULPA Filter Efficiency		>99.999% at 0.1 to 0.3 micron as per IEST-RP-CC001 USA	
		>99.999% at MPPS as per EN 1822 (H14) EU	
Sound Emission**	NSF / ANSI 49	61.3 dBA	62.5 dBA
	EN 12469	58.3 dBA	59.5 dBA
Fluorescent Lamp Intensity (lux)		1400	
Fluorescent Lamp Intensity (ft-cd)		130	
Cabinet Construction	Main Body	"1.2 mm (0.05") 18-gauge electro-galvanized steel with white oven-baked epoxy-polyester Isocide antimicrobial powder coated finish"	
	Work Zone	1.5 mm (0.06") 16-gauge Stainless steel Type 304 with No. 4B finish	
	Side Walls (G-Series)	UV absorbing tempered glass, 5 mm (0.2") colourless and transparent	
	Side Walls (D-Series)	1.5 mm (0.06") 16 gauge stainless steel, type 304, with 4B finish	
Electrical	Cabinet Full Load Amps (FLA)	9.4 A	12.6 A
	Heat load (BTU/Hr)	905	1230
Nominal Power Consumption		265 W	360 W
Net Weight***		240 Kg (529 lbs)	366 Kg (807 lbs)
Shipping Weight***		295 Kg (650 lbs)	448 Kg (988 lbs)
Shipping Dimensions, Maximum (W x D x H) mm ***		1450 x 820 x 1760	2050 x 820 x 1760
Shipping Volume, Maximum***		2.09 m ³	2.96 m ³

* Depth includes the remove-able arm rest and front cover. When they are removed, depth is 790 mm (31.1").

** Noise reading in open field / anechoic chamber.

*** Cabinet only, excludes optional stand

Manual Revision History

[illegible]

